1 Measures

Section 1.1 deals with algebraic operations on subsets of a given nonempty set Ω. Section 1.2 treats nonnegative set functions on classes of sets and defines the notion of a measure on an algebra. Section 1.3 treats the extension theorem, and Section 1.4 deals with completeness of measures.

1.1 Classes of sets

Let Ω be a nonempty set and $\mathcal{P}(\Omega) \equiv \{A : A \subset \Omega\}$ be the *power set of* Ω, i.e., the class of all subsets of Ω.

Definition 1.1.1: A collection of sets $\mathcal{F} \subset \mathcal{P}(\Omega)$ is called an *algebra* if (a) $\Omega \in \mathcal{F}$, (b) $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$, and (c) $A, B \in \mathcal{F}$ implies $A \cup B \in \mathcal{F}$ (i.e., closure under pairwise unions).

Thus, an algebra is a class of sets containing Ω that is closed under complementation and pairwise (and hence finite) unions. It is easy to see that one can equivalently define an algebra by requiring that properties (a), (b) hold and that the property

$$\text{(c)}' \quad A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$$

holds (i.e. closure under finite intersections).

Definition 1.1.2: A class $\mathcal{F} \subset \mathcal{P}(\Omega)$ is called a *σ-algebra* if it is an algebra and if it satisfies

$$(d) \quad A_n \in \mathcal{F} \quad \text{for} \quad n \geq 1 \Rightarrow \bigcup_{n \geq 1} A_n \in \mathcal{F}.$$

Thus, a σ-algebra is a class of subsets of Ω that contains Ω and is closed under complementation and countable unions. As pointed out in the introductory chapter, a σ-algebra can be alternatively defined as an algebra that is closed under monotone unions as the following shows.

Proposition 1.1.1: *Let $\mathcal{F} \subset \mathcal{P}(\Omega)$. Then $\mathcal{F}$ is a σ-algebra if and only if $\mathcal{F}$ is an algebra and satisfies*

$$A_n \in \mathcal{F}, A_n \subset A_{n+1} \quad \textit{for all} \quad n \Rightarrow \bigcup_{n \geq 1} A_n \in \mathcal{F}.$$

Proof: The 'only if' part is obvious. For the 'if' part, let $\{B_n\}_{n=1}^{\infty} \subset \mathcal{F}$. Then, since $\mathcal{F}$ is an algebra, $A_n \equiv \bigcup_{j=1}^{n} B_j \in \mathcal{F}$ for all n. Further, $A_n \subset A_{n+1}$ for all n and $\bigcup_{n \geq 1} B_n = \bigcup_{n \geq 1} A_n$. Since by hypothesis $\cup_n A_n \in \mathcal{F}$, $\cup_n B_n \in \mathcal{F}$. □

Here are some examples of algebras and σ-algebras.

Example 1.1.1: Let $\Omega = \{a, b, c, d\}$. Consider the classes

$$\mathcal{F}_1 = \{\Omega, \emptyset, \{a\}\}$$

and

$$\mathcal{F}_2 = \{\Omega, \emptyset, \{a\}, \{b, c, d\}\}.$$

Then, $\mathcal{F}_2$ is an algebra (and also a σ-algebra), but $\mathcal{F}_1$ is not an algebra, since $\{a\}^c \notin \mathcal{F}_1$.

Example 1.1.2: Let Ω be any nonempty set and let

$$\mathcal{F}_3 = \mathcal{P}(\Omega) \equiv \{A : A \subset \Omega\}, \quad \text{the power set of} \quad \Omega$$

and

$$\mathcal{F}_4 = \{\Omega, \emptyset\}.$$

Then, it is easy to check that both $\mathcal{F}_3$ and $\mathcal{F}_4$ are σ-algebras. The latter σ-algebra is often called the *trivial σ-algebra* on Ω (Problem 1.1).

From the definition it is clear that any σ-algebra is also an algebra and thus $\mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4$ are examples of algebras, too. The following is an example of an algebra that is not a σ-algebra.

Example 1.1.3: Let Ω be a nonempty set, and let $|A|$ denote the number of elements of a set $A \subset \Omega$. Define.

$$\mathcal{F}_5 = \{A \subset \Omega : \text{ either } |A| \text{ is finite or } |A^c| \text{ is finite}\}.$$

Then, note that (i) $\Omega \in \mathcal{F}_5$ (since $|\Omega^c| = |\emptyset| = 0$)), (ii) $A \in \mathcal{F}_5$ implies $A^c \in \mathcal{F}_5$ (if $|A| < \infty$, then $|(A^c)^c| = |A| < \infty$ and if $|A^c| < \infty$, then $A^c \in \mathcal{F}_5$ trivially). Next, suppose that $A, B \in \mathcal{F}_5$. If either $|A| < \infty$ or $|B| < \infty$, then

$$|A \cap B| \leq \min\{|A|, |B|\} < \infty,$$

so that $A \cap B \in \mathcal{F}_5$. On the other hand, if both $|A^c| < \infty$ and $|B^c| < \infty$, then

$$|(A \cap B)^c| = |A^c \cup B^c| \leq |A^c| + |B^c| < \infty,$$

implying that $A \cap B \in \mathcal{F}_5$. Thus, property (c)$'$ holds, and $\mathcal{F}_5$ is an algebra. However, if $|\Omega| = \infty$, then $\mathcal{F}_5$ is not a σ-algebra. To see this, suppose that $|\Omega| = \infty$ and $\{\omega_1, \omega_2, \ldots\} \subset \Omega$. Then, by definition, $A_i = \{\omega_i\} \in \mathcal{F}_5$ for all $i \geq 1$, but $A \equiv \bigcup_{i=1}^{\infty} A_{2i-1} = \{\omega_1, \omega_3, \ldots\} \notin \mathcal{F}_5$, since $|A| = |A^c| = \infty$.

Example 1.1.4: Let Ω be a nonempty set and let

$$\mathcal{F}_6 = \{A \subset \Omega : A \text{ is countable or } A^c \text{ is countable}\}.$$

Then, it is easy to show that $\mathcal{F}_6$ is a σ-algebra (Problem 1.3).

Suppose $\{\mathcal{F}_\theta : \theta \in \Theta\}$ is a family of σ-algebras on Ω. From the definition, it follows that the intersection $\bigcap_{\theta \in \Theta} \mathcal{F}_\theta$ is a σ-algebra, no matter how large the index set Θ is (Problem 1.4). However, the union of two σ-algebras may not even be an algebra (Problem 1.5). For the development of measure theory and probability theory, the concept of a σ-algebra plays a crucial role. In many instances, given an arbitrary collection of subsets of Ω, one would like to extend it to a possibly larger class that is a σ-algebra. This leads to the following definition.

Definition 1.1.3: If $\mathcal{A}$ is a class of subsets of Ω, then the σ-algebra generated by $\mathcal{A}$, denoted by $\sigma\langle\mathcal{A}\rangle$, is defined as

$$\sigma\langle\mathcal{A}\rangle = \bigcap_{\mathcal{F} \in \mathcal{I}(\mathcal{A})} \mathcal{F},$$

where $\mathcal{I}(\mathcal{A}) \equiv \{\mathcal{F} : \mathcal{A} \subset \mathcal{F} \text{ and } \mathcal{F} \text{ is a } \sigma\text{-algebra on } \Omega\}$ is the collection of all σ-algebras containing the class $\mathcal{A}$.

Note that since the power set $\mathcal{P}(\Omega)$ contains $\mathcal{A}$ and is itself a σ-algebra, the collection $\mathcal{I}(\mathcal{A})$ is not empty and hence, the intersection in the above definition is well defined.

Example 1.1.5: In the setup of Example 1.1.1, $\sigma\langle\mathcal{F}_1\rangle = \mathcal{F}_2$ (why?).

A particularly useful class of σ-algebras are those generated by open sets of a topological space. These are called Borel σ-algebras. A *topological space* is a pair $(\mathbb{S}, \mathcal{T})$ where $\mathbb{S}$ is a nonempty set and $\mathcal{T}$ is a collection of subsets of $\mathbb{S}$ such that (i) $\mathbb{S} \in \mathcal{T}$, (ii) $\mathcal{O}_1, \mathcal{O}_2 \in \mathcal{T} \Rightarrow \mathcal{O}_1 \cap \mathcal{O}_2 \in \mathcal{T}$, and (iii) $\{\mathcal{O}_\alpha : \alpha \in I\} \subset \mathcal{T} \Rightarrow \bigcup_{\alpha \in I} \mathcal{O}_\alpha \in \mathcal{T}$. Elements of $\mathcal{T}$ are called *open sets*.

A *metric space* is a pair $(\mathbb{S}, d)$ where $\mathbb{S}$ is a nonempty set and d is a function from $\mathbb{S} \times \mathbb{S}$ to $\mathbb{R}^+$ satisfying (i) $d(x,y) = d(y,x)$ for all x, y in $\mathbb{S}$, (ii) $d(x,y) = 0$ iff $x = y$, and (iii) $d(x,z) \leq d(x,y) + d(y,z)$ for all x, y, z in $\mathbb{S}$. Property (iii) is called the triangle inequality. The function d is called a metric on $\mathbb{S}$ (cf. see A.4).

Any Euclidean space $\mathbb{R}^n (1 \leq n < \infty)$ is a metric space under any one of the following metrics:

(a) For $1 \leq p < \infty$, $d_p(x,y) = \Big(\sum_{i=1}^{n} |x_i - y_i|^p \Big)^{1/p}$.

(b) $d_\infty(x,y) = \max_{1 \leq i \leq n} |x_i - y_i|$.

(c) For $0 < p < 1$, $d_p(x,y) = \Big(\sum_{i=1}^{n} |x_i - y_i|^p \Big)$.

A metric space $(\mathbb{S}, d)$ is a topological space where a set $\mathcal{O}$ is open if for all $x \in \mathcal{O}$, there is an $\epsilon > 0$ such that $B(x, \epsilon) \equiv \{y : d(y,x) < \epsilon\} \subset \mathcal{O}$.

Definition 1.1.4: The *Borel σ-algebra* on a topological space $\mathbb{S}$ (in particular, on a metric space or an Euclidean space) is defined as the σ-algebra generated by the collection of open sets in $\mathbb{S}$.

Example 1.1.6: Let $\mathcal{B}(\mathbb{R}^k)$ denote the Borel σ-algebra on $\mathbb{R}^k$, $1 \leq k < \infty$. Then,

$$\mathcal{B}(\mathbb{R}^k) \equiv \sigma\langle \{A : A \text{ is an open subset of } \mathbb{R}^k\} \rangle$$

is also generated by each of the following classes of sets

$$\begin{aligned}
\mathcal{O}_1 &= \{(a_1, b_1) \times \ldots \times (a_k, b_k) : -\infty \leq a_i < b_i \leq \infty, 1 \leq i \leq k\}; \\
\mathcal{O}_2 &= \{(-\infty, x_1) \times \cdots \times (-\infty, x_k) : x_1, \ldots, x_k \in \mathbb{R}\}; \\
\mathcal{O}_3 &= \{(a_1, b_1) \times \ldots \times (a_k, b_k) : a_i, b_i \in \mathcal{Q}, a_i < b_i, 1 \leq i \leq k\}; \\
\mathcal{O}_4 &= \{(-\infty, x_1) \times \ldots \times (-\infty, x_k) : x_1, \ldots, x_k \in \mathcal{Q}\},
\end{aligned}$$

where $\mathcal{Q}$ denotes the set of all rational numbers.

To show this, note that $\sigma\langle \mathcal{O}_i \rangle \subset \mathcal{B}(\mathbb{R}^k)$ for $i = 1, 2, 3, 4$, and hence, it is enough to show that $\sigma\langle \mathcal{O}_i \rangle \supset \mathcal{B}(\mathbb{R}^k)$. Let $\mathcal{G}$ be a σ-algebra containing $\mathcal{O}_3$. Observe that given any open set $A \subset \mathbb{R}^k$, there exist a sequence of sets $\{B_n\}_{n \geq 1}$ in $\mathcal{O}_3$ such that $A = \bigcup_{n \geq 1} B_n$ (Problem 1.9). Since $\mathcal{G}$ is a σ-algebra and $B_n \in \mathcal{G}$ for all $n \geq 1$, $A \in \mathcal{G}$. Thus, $\mathcal{G}$ is a σ-algebra containing all open subsets of $\mathbb{R}^k$, and hence $\mathcal{G} \supset \mathcal{B}(\mathbb{R}^k)$. Hence, it follows that

$$\mathcal{B}(\mathbb{R}^k) \supset \sigma\langle \mathcal{O}_1 \rangle \supset \sigma\langle \mathcal{O}_3 \rangle = \bigcap_{\mathcal{G}: \mathcal{G} \supset \mathcal{O}_3} \mathcal{G} \supset \mathcal{B}(\mathbb{R}^k).$$

Next note that any interval $(a,b) \subset \mathbb{R}$ can be expressed in terms of half spaces of the form $(-\infty, x),\ x \in \mathbb{R}$ as

$$(a,b) = \bigcup_{n=1}^{\infty} [(-\infty, b) \setminus (-\infty, a + n^{-1})],$$

where for any two sets A and B, $A \backslash B = \{x : x \in A, x \notin B\}$. It is not difficult to show that this implies that $\sigma\langle \mathcal{O}_i \rangle = \mathcal{B}(\mathbb{R}^k)$ for $i = 2, 4$ (Problem 1.10).

Example 1.1.7: Let Ω be a nonempty set with $|\Omega| = \infty$ and $\mathcal{F}_5$ and $\mathcal{F}_6$ be as in Examples 1.1.3 and 1.1.4. Then $\mathcal{F}_6 = \sigma\langle \mathcal{F}_5 \rangle$. To see this, note that $\mathcal{F}_6$ is a σ-algebra containing $\mathcal{F}_5$, so that $\sigma\langle \mathcal{F}_5 \rangle \subset \mathcal{F}_6$. To prove the reverse inclusion, let $\mathcal{G}$ be a σ-algebra containing $\mathcal{F}_5$. It is enough to show that $\mathcal{F}_6 \subset \mathcal{G}$. Let $A \in \mathcal{F}_6$. If A is countable, say $A = \{\omega_1, \omega_2, \ldots\}$, then $A_i \equiv \{\omega_i\} \in \mathcal{F}_5 \subset \mathcal{G}$ for all $i \geq 1$ and hence $A = \bigcup_{i=1}^{\infty} A_i \in \mathcal{G}$. On the other hand, if A^c is countable, then by the above argument, $A^c \in \mathcal{G} \Rightarrow A \in \mathcal{G}$.

Definition 1.1.5: A class $\mathcal{C}$ of subsets of Ω is a π-*system* or a π-*class* if $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$.

Example 1.1.8: The class $\mathcal{C}$ of intervals in $\mathbb{R}$ is a π-system whereas the class of all open discs in $\mathbb{R}^2$ is not.

Definition 1.1.6: A class $\mathcal{L}$ of subsets of Ω is a λ-*system* or a λ-*class* if (i) $\Omega \in \mathcal{L}$, (ii) $A, B \in \mathcal{L}, A \subset B \Rightarrow B \setminus A \in \mathcal{L}$, and (iii) $A_n \in \mathcal{L}, A_n \subset A_{n+1}$ for all $n \geq 1 \ \Rightarrow \bigcup_{n \geq 1} A_n \in \mathcal{L}$.

Example 1.1.9: Every σ-algebra $\mathcal{F}$ is a λ-system. But an algebra need not be a λ-system.

It is easily checked that if $\mathcal{L}_1$ and $\mathcal{L}_2$ are λ-systems, then $\mathcal{L}_1 \cap \mathcal{L}_2$ is also a λ-system. Recall that $\sigma\langle \mathcal{B} \rangle$, the σ-algebra generated by $\mathcal{B}$, is the intersection of all σ-algebras containing $\mathcal{B}$ and is also the smallest σ-algebra containing $\mathcal{B}$. Similarly, for any $\mathcal{B} \subset \mathcal{P}(\Omega)$, the λ-system generated by $\mathcal{B}$, denoted by $\lambda\langle \mathcal{B} \rangle$, is defined as the intersection of all λ-systems containing $\mathcal{B}$. It is the smallest λ-system containing $\mathcal{B}$.

Theorem 1.1.2: (*The* π-λ *theorem*). *If* $\mathcal{C}$ *is a* π-*system, then* $\lambda\langle \mathcal{C} \rangle = \sigma\langle \mathcal{C} \rangle$.

Proof: For any $\mathcal{C}$, $\sigma\langle \mathcal{C} \rangle$ is a λ-system and $\sigma\langle \mathcal{C} \rangle$ contains $\mathcal{C}$. Thus, $\lambda\langle \mathcal{C} \rangle \subset \sigma\langle \mathcal{C} \rangle$ for any $\mathcal{C}$. Hence, it suffices to show that if $\mathcal{C}$ is a π-system, then $\lambda\langle \mathcal{C} \rangle$ is a σ-algebra . Since $\lambda\langle \mathcal{C} \rangle$ is a λ-system, it is closed under complementation and monotone increasing unions. By Proposition 1.1.1, it is enough to show that it is closed under intersection. Let $\lambda_1(\mathcal{C}) \equiv \{A : A \in \lambda\langle \mathcal{C} \rangle, A \cap B \in \lambda\langle \mathcal{C} \rangle \text{ for all } B \in \mathcal{C}\}$. Then, $\lambda_1(\mathcal{C})$ is a λ-system and $\mathcal{C}$ being a π-system, $\lambda_1(\mathcal{C}) \supset \mathcal{C}$. Therefore, $\lambda_1(\mathcal{C}) \supset \lambda\langle \mathcal{C} \rangle$. But $\lambda_1(\mathcal{C}) \subset \lambda\langle \mathcal{C} \rangle$. So $\lambda_1(\mathcal{C}) = \lambda\langle \mathcal{C} \rangle$.

Next, let $\lambda_2(\mathcal{C}) \equiv \{A : A \in \lambda\langle\mathcal{C}\rangle, A \cap B \in \lambda\langle\mathcal{C}\rangle \text{ for all } B \in \lambda\langle\mathcal{C}\rangle\}$. Then $\lambda_2(\mathcal{C})$ is a λ-system and by the previous step $\mathcal{C} \subset \lambda_2(\mathcal{C}) \subset \lambda\langle\mathcal{C}\rangle$. Hence, it follows that $\lambda_2(\mathcal{C}) = \lambda\langle\mathcal{C}\rangle$, i.e., $\lambda\langle\mathcal{C}\rangle$ is closed under intersection. This completes the proof of the theorem. $\square$

Corollary 1.1.3: *If $\mathcal{C}$ is a π-system and $\mathcal{L}$ is a λ-system containing $\mathcal{C}$, then $\mathcal{L} \supset \sigma\langle\mathcal{C}\rangle$.*

Remark 1.1.1: There are several equivalent definitions of λ-systems; see, for example, Billingsley (1995). A closely related concept is that of a monotone class; see, for example, Chung (1974).

1.2 Measures

A *set function* is an extended real valued function defined on a class of subsets of a set Ω. Measures are nonnegative set functions that, intuitively speaking, measure the content of a subset of Ω. As explained in Section 2 of the introductory chapter, a measure has to satisfy certain natural requirements, such as the measure of the union of a countable collection of *disjoint* sets is the *sum* of the measures of the individual sets. Formally, one has the following definition.

Definition 1.2.1: Let Ω be a nonempty set and $\mathcal{F}$ be an algebra on Ω. Then, a set function μ on $\mathcal{F}$ is called a *measure* if

(a) $\mu(A) \in [0, \infty]$ for all $A \in \mathcal{F}$;

(b) $\mu(\emptyset) = 0$;

(c) for any disjoint collection of sets $A_1, A_2, \ldots, \in \mathcal{F}$ with $\bigcup_{n \geq 1} A_n \in \mathcal{F}$,

$$\mu\Big(\bigcup_{n \geq 1} A_n\Big) = \sum_{n=1}^{\infty} \mu(A_n).$$

As discussed in Section 2 of the introductory chapter, these conditions on μ are equivalent to finite additivity and monotone continuity from below.

Proposition 1.2.1: *Let Ω be a nonempty set and $\mathcal{F}$ be an algebra of subsets of Ω and μ be a set function on $\mathcal{F}$ with values in $[0, \infty]$ and with $\mu(\emptyset) = 0$. Then, μ is a measure iff μ satisfies*

$(iii)'_a$: *(finite additivity) for all $A_1, A_2 \in \mathcal{F}$ with $A_1 \cap A_2 = \emptyset$, $\mu(A_1 \cup A_2) = \mu(A_1) + \mu(A_2)$, and*

$(iii)'_b$: *(monotone continuity from below or, m.c.f.b., in short) for any collection $\{A_n\}_{n \geq 1}$ of sets in $\mathcal{F}$ such that $A_n \subset A_{n+1}$ for all $n \geq 1$*

and $\bigcup_{n\geq 1} A_n \in \mathcal{F}$,

$$\mu\Big(\bigcup_{n\geq 1} A_n\Big) = \lim_{n\to\infty} \mu(A_n).$$

Proof: Let μ be a measure on $\mathcal{F}$. Since μ satisfies (iii), taking $A_3, A_4, \ldots$ to be $\emptyset$ yields $(\text{iii})'_a$. This implies that for A and B in $\mathcal{F}$, $A \subset B \Rightarrow \mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$, i.e., μ is *monotone*. To establish $(\text{iii})'_b$, note that if $\mu(A_n) = \infty$ for some $n = n_0$, then $\mu(A_n) = \infty$ for all $n \geq n_0$ and $\mu(\bigcup_{n\geq 1} A_n) = \infty$ and $(\text{iii})'_b$ holds in this case. Hence, suppose that $\mu(A_n) < \infty$ for all $n \geq 1$. Setting $B_n = A_n \setminus A_{n-1}$ for $n \geq 1$ (with $A_0 = \emptyset$), by $(\text{iii})'_a$, $\mu(B_n) = \mu(A_n) - \mu(A_{n-1})$. Since $\{B_n\}_{n\geq 1}$ is a *disjoint* collection of sets in $\mathcal{F}$ with $\bigcup_{n\geq 1} B_n = \bigcup_{n\geq 1} A_n$, by (iii)

$$\begin{aligned}\mu\Big(\bigcup_{n\geq 1} A_n\Big) &= \mu\Big(\bigcup_{n\geq 1} B_n\Big) = \sum_{n=1}^{\infty} \mu(B_n) = \lim_{N\to\infty} \sum_{n=1}^{N} [\mu(A_n) - \mu(A_{n-1})] \\ &= \lim_{N\to\infty} \mu(A_N),\end{aligned}$$

and so $(\text{iii})'_b$ holds also in this case.

Conversely, let μ satisfy $\mu(\emptyset) = 0$ and $(\text{iii})'_a$ and $(\text{iii})'_b$. Let $\{A_n\}_{n\geq 1}$ be a disjoint collection of sets in $\mathcal{F}$ with $\bigcup_{i\geq 1} A_i \in \mathcal{F}$. Let $C_n = \bigcup_{j=1}^{n} A_j$ for $n \geq 1$. Since $\mathcal{F}$ is an algebra, $C_n \in \mathcal{F}$ for all $n \geq 1$. Also, $C_n \subset C_{n+1}$ for all $n \geq 1$. Hence, $\bigcup_{n\geq 1} C_n = \bigcup_{j\geq 1} A_j$. By $(\text{iii})'_b$,

$$\begin{aligned}\mu\Big(\bigcup_{j\geq 1} A_j\Big) &= \mu\Big(\bigcup_{n\geq 1} C_n\Big) = \lim_{n\to\infty} \mu(C_n) \\ &= \lim_{n\to\infty} \sum_{j=1}^{n} \mu(A_j) \quad (\text{by } (\text{iii})'_a) \\ &= \sum_{j=1}^{\infty} \mu(A_j).\end{aligned}$$

Thus, (iii) holds. □

Remark 1.2.1: The definition of a measure given in Definition 1.2.1 is valid when $\mathcal{F}$ is a σ-algebra. However, very often one may start with a measure on an algebra $\mathcal{A}$ and then extend it to a measure on the σ-algebra $\sigma\langle\mathcal{A}\rangle$. This is why the definition of a measure on an algebra is given here. In the same vein, one may begin with a definition of a measure on a class of subsets of Ω that form only a semialgebra (cf. Definition 1.3.1). As described in the introductory chapter, such preliminary collections of sets are "nice" sets for which the measure may be defined easily, and the extension to a σ-algebra containing these sets may be necessary if one is interested in more general sets. This topic is discussed in greater detail in the next section.

Definition 1.2.2: A measure μ is called *finite* or *infinite* according as $\mu(\Omega) < \infty$ or $\mu(\Omega) = \infty$. A finite measure with $\mu(\Omega) = 1$ is called a *probability* measure. A measure μ on a σ-algebra $\mathcal{F}$ is called *σ-finite* if there exist a countable collection of sets $A_1, A_2, \ldots, \in \mathcal{F}$, not necessarily disjoint, such that

$$\text{(a)} \bigcup_{n \geq 1} A_n = \Omega \quad \text{and} \quad \text{(b)}\ \mu(A_n) < \infty \quad \text{for all} \quad n \geq 1.$$

Here are some examples of measures.

Example 1.2.1: (*The counting measure*). Let Ω be a nonempty set and $\mathcal{F}_3 = \mathcal{P}(\Omega)$ be the set of all subsets of Ω (cf. Example 1.1.2). Define

$$\mu(A) = |A|, \quad A \in \mathcal{F}_3,$$

where $|A|$ denotes the number of elements in A. It is easy to check that μ satisfies the requirements (a)–(c) of a measure. This measure μ is called the *counting measure* on Ω. Note that μ is finite iff Ω is finite and it is σ-finite if Ω is countably infinite.

Example 1.2.2: (*Discrete probability measures*). Let $\omega_1, \omega_2, \ldots, \in \Omega$ and $p_1, p_2, \ldots \in [0, 1]$ be such that $\sum_{i=1}^\infty p_i = 1$. Define for any $A \subset \Omega$

$$P(A) = \sum_{i=1}^\infty p_i I_A(\omega_i),$$

where $I_A(\cdot)$ denotes the indicator function of a set A, defined by $I_A(\omega) = 0$ or 1 according as $\omega \notin A$ or $\omega \in A$. For any disjoint collection of sets $A_1, A_2, \ldots \in \mathcal{P}(\Omega)$,

$$\begin{aligned} P\Big(\bigcup_{i=1}^\infty A_i\Big) &= \sum_{j=1}^\infty p_j I_{\bigcup_{i=1}^\infty A_i}(\omega_j) \\ &= \sum_{j=1}^\infty p_j \Big(\sum_{i=1}^\infty I_{A_i}(\omega_j)\Big) \\ &= \sum_{i=1}^\infty \Big(\sum_{j=1}^\infty p_j I_{A_i}(\omega_j)\Big) \\ &= \sum_{i=1}^\infty P(A_i), \end{aligned}$$

where interchanging the order of summation is permissible since the summands are nonnegative. This shows that P is a probability measure on $\mathcal{P}(\Omega)$.

Example 1.2.3: (*Lebesgue-Stieltjes measures on* $\mathbb{R}$). As mentioned in the previous chapter (cf. Section 2), a large class of measures on the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ of subsets of $\mathbb{R}$, known as the *Lebesgue-Stieltjes measures*, arise from nondecreasing right continuous functions $F : \mathbb{R} \to \mathbb{R}$. For each such F, the corresponding measure μ_F satisfies $\mu_F((a, b]) = F(b) - F(a)$ for all $-\infty < a < b < \infty$. The construction of these μ_F's via the extension theorem will be discussed in the next section. Also, note that if $A_n = (-n, n)$, $n = 1, 2, \ldots$, then $\mathbb{R} = \bigcup_{n \geq 1} A_n$ and $\mu_F(A_n) < \infty$ for each $n \geq 1$ (such measures are called Radon measures) and thus, μ_F is necessarily σ-finite.

Proposition 1.2.2: *Let μ be a measure on an algebra $\mathcal{F}$, and let $A, B, A_1, \ldots, A_k \in \mathcal{F}, 1 \leq k < \infty$. Then,*

(i) (*Monotonicity*) $\mu(A) \leq \mu(B)$ *if* $A \subset B$;

(ii) (*Finite subadditivity*) $\mu(A_1 \cup \ldots \cup A_k) \leq \mu(A_1) + \ldots + \mu(A_k)$;

(iii) (*Inclusion-exclusion formula*) *If* $\mu(A_i) < \infty$ *for all* $i = 1, \ldots, k$, *then*

$$\begin{aligned} \mu(A_1 \cup \ldots \cup A_k) &= \sum_{i=1}^{k} \mu(A_i) - \sum_{1 \leq i < j < k} \mu(A_i \cap A_j) \\ &\quad + \ldots + (-1)^{k-1} \mu(A_1 \cap \ldots \cap A_k). \end{aligned}$$

Proof: $\mu(B) = \mu(A \cup (A^c \cap B)) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$, by (a) and (c) of Definition 1.2.1. This proves (i).

To prove (ii), note that if either $\mu(A)$ or $\mu(B)$ is finite, then $\mu(A \cap B) < \infty$, by (i). Hence, using the countable additivity property (c), we have

$$\begin{aligned} \mu(A \cup B) &= \mu(A) + \mu(B \setminus A) \\ &= \mu(A) + [\mu(B \setminus A) + \mu(A \cap B)] - \mu(A \cap B) \\ &= \mu(A) + \mu(B) - \mu(A \cap B). \end{aligned} \tag{2.1}$$

Hence, (ii) follows from (2.1) and by induction.

To prove (iii), note that the case $k = 2$ follows from (2.1). Next, suppose that (iii) holds for all sets $A_1, \ldots, A_k \in \mathcal{F}$ with $\mu(A_i) < \infty$ for all $i = 1, \ldots, k$ for some $k = n$, $n \in \mathbb{N}$. To show that it holds for $k = n + 1$, note that by (2.1),

$$\begin{aligned} &\mu\Big(\bigcup_{i=1}^{n+1} A_i\Big) \\ = \;& \mu\Big(\bigcup_{i=1}^{n} A_i\Big) + \mu(A_{n+1}) - \mu\Big(\bigcup_{i=1}^{n} (A_i \cap A_{n+1})\Big) \end{aligned}$$

$$
\begin{aligned}
&= \Big\{ \sum_{i=1}^{n} \mu(A_i) - \sum_{1<i<j\le n} \mu(A_i \cap A_j) + \cdots + (-1)^{n-1}\mu(A_1 \cap \ldots \cap A_n) \Big\} \\
&\quad + \mu(A_{n+1}) - \Big[\sum_{i=1}^{n} \mu(A_i \cap A_{n+1}) - \sum_{1\le i<j\le n} \mu(A_i \cap A_j \cap A_{n+1}) \\
&\qquad\qquad + \cdots + (-1)^{n-1}\mu(A_1 \cap \ldots \cap A_{n+1}) \Big] \\
&= \sum_{i=1}^{n+1} \mu(A_i) - \sum_{1\le i<j\le n+1} \mu(A_i \cap A_j) + \cdots + (-1)^{n}\mu\Big(\bigcap_{j=1}^{n+1} A_j \Big).
\end{aligned}
$$

By induction, this completes the proof of Proposition 1.2.2. □

In Proposition 1.2.1, it was shown that a set function μ on an algebra $\mathcal{F}$ is a measure iff it is finitely additive and monotone continuous from below. A natural question is: if μ is a measure on $\mathcal{F}$ and $\{A_n\}_{n\ge1}$ is a collection of decreasing sets in $\mathcal{F}$ with $A \equiv \bigcap_{n\ge1} A_n$ also in $\mathcal{F}$, does the relation $\mu(A) = \lim_{n\to\infty} \mu(A_n)$ hold, i.e., does *monotone continuity from above* hold? The answer is positive under the assumption $\mu(A_{n_0}) < \infty$ for some $n_0 \in \mathbb{N}$. It turns out that, in general, this assumption cannot be dropped (Problem 1.18).

Proposition 1.2.3: *Let μ be a measure on an algebra $\mathcal{F}$.*

(i) (*Monotone continuity from above*) *Let $\{A_n\}_{n\ge1}$ be a sequence of sets in $\mathcal{F}$ such that $A_{n+1} \subset A_n$ for all $n \ge 1$ and $A \equiv \bigcap_{n\ge1} A_n \in \mathcal{F}$. Also, let $\mu(A_{n_0}) < \infty$ for some $n_0 \in \mathbb{N}$. Then,*

$$\lim_{n\to\infty} \mu(A_n) = \mu(A).$$

(ii) (*Countable subadditivity*) *If $\{A_n\}_{n\ge1}$ is a sequence of sets in $\mathcal{F}$ such that $\bigcup_{n\ge1} A_n \in \mathcal{F}$, then*

$$\mu\Big(\bigcup_{n=1}^{\infty} A_n \Big) \le \sum_{n=1}^{\infty} \mu(A_n).$$

Proof: To prove part (i), without loss of generality (w.l.o.g.), assume that $n_0 = 1$, i.e., $\mu(A_1) < \infty$. Let $C_n = A_1 \setminus A_n$ for $n \ge 1$, and $C_\infty = A_1 \setminus A$. Then C_n and C_∞ belong to $\mathcal{F}$ and $C_n \uparrow C_\infty$. By Proposition 1.2.1 $(\text{iii})_b'$, (i.e., by the m.c.f.b. property), $\mu(C_n) \uparrow \mu(C_\infty)$ and by $(\text{iii})_a'$, (i.e., finite additivity), $\mu(C_n) = \mu(A_1) - \mu(A_n)$ for all $1 \le n < \infty$, due to the fact $\mu(A_1) < \infty$. This proves (i).

To prove part (ii), let $D_n = \bigcup_{i=1}^n A_i, n \geq 1$. Then, $D_n \uparrow D \equiv \bigcup_{i \geq 1} A_i$. Hence, by m.c.f.b. and finite subadditivity,

$$\mu(D) = \lim_{n\to\infty} \mu(D_n) \leq \lim_{n\to\infty} \sum_{i=1}^{n} \mu(A_i) = \sum_{n=1}^{\infty} \mu(A_n). \qquad \square$$

Theorem 1.2.4: (*Uniqueness of measures*). *Let μ_1 and μ_2 be two finite measures on a measurable space $(\Omega, \mathcal{F})$. Let $\mathcal{C} \subset \mathcal{F}$ be a π-system such that $\mathcal{F} = \sigma\langle\mathcal{C}\rangle$. If $\mu_1(C) = \mu_2(C)$ for all $C \in \mathcal{C}$ and $\mu_1(\Omega) = \mu_2(\Omega)$, then $\mu_1(A) = \mu_2(A)$ for all $A \in \mathcal{F}$.*

Proof: Let $\mathcal{L} \equiv \{A : A \in \mathcal{F},\ \mu_1(A) = \mu_2(A)\}$. It is easy to verify that $\mathcal{L}$ is a λ-system. Since $\mathcal{C} \subset \mathcal{L}$, by Theorem 1.1.2, $\mathcal{L} = \sigma\langle\mathcal{C}\rangle = \mathcal{F}$. $\square$

1.3 The extension theorems and Lebesgue-Stieltjes measures

As discussed earlier, in many situations, one starts with a given set function μ defined on a small class $\mathcal{C}$ of subsets of a set Ω and then wants to extend μ to a larger class $\mathcal{M}$ by some approximation procedure. In this section, a general result in this direction, known as the *extension theorem*, is established. This is then applied to the construction of Lebesgue-Stieltjes measures on Euclidean spaces. For another application, see Chapter 6.

1.3.1 Caratheodory extension of measures

Definition 1.3.1: Let Ω be a nonempty set and let $\mathcal{P}(\Omega)$ be the power set of Ω. A class $\mathcal{C} \subset \mathcal{P}(\Omega)$ is called a *semialgebra* if (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$ and (ii) for any $A \in \mathcal{C}$, there exist sets $B_1, B_2, \ldots, B_k \in \mathcal{C}$, for some $1 \leq k < \infty$, such that $B_i \cap B_j = \emptyset$ for $i \neq j$, and $A^c = \bigcup_{i=1}^k B_i$.

Example 1.3.1: $\Omega = \mathbb{R},\ \mathcal{C} \equiv \{(a, b], (b, \infty) : -\infty \leq a, b < \infty\}$.

Example 1.3.2: $\Omega = \mathbb{R},\ \mathcal{C} \equiv \{I : I \text{ is an interval}\}$. An interval I in $\mathbb{R}$ is a set in $\mathbb{R}$ such that $a,\ b \in I,\ a < b \Rightarrow (a, b) \subset I$.

Example 1.3.3: $\Omega = \mathbb{R}^k$, $\mathcal{C} \equiv \{I_1 \times I_2 \times \ \times I_k : I_j$ is an interval in $\mathbb{R}$ for $1 \leq j \leq k\}$.

Recall that a collection $\mathcal{A} \subset \mathcal{P}(\Omega)$ is an algebra if it is closed under finite union and complementation. It is easily verified (Problem 1.19) that the smallest algebra containing a semialgebra $\mathcal{C}$ is $\mathcal{A}(\mathcal{C}) \equiv \{A : A = \bigcup_{i=1}^k B_i, B_i \in \mathcal{C}$ for $i = 1, \ldots, k, k < \infty, \}$, i.e., the class of finite unions of sets from $\mathcal{C}$.

Definition 1.3.2: A set function μ on a semialgebra $\mathcal{C}$, taking values in $\bar{\mathbb{R}}_+ \equiv [0,\infty]$, is called a *measure* if (i) $\mu(\emptyset) = 0$ and (ii) for any sequence of sets $\{A_n\}_{n\geq 1} \subset \mathcal{C}$ with $\bigcup_{n\geq 1} A_n \in \mathcal{C}$, and $A_i \cap A_j = \emptyset$ for $i \neq j$, $\mu(\bigcup_{n\geq 1} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$.

Proposition 1.3.1: *Let μ be a measure on a semialgebra $\mathcal{C}$. Let $\mathcal{A} \equiv \mathcal{A}(\mathcal{C})$ be the smallest algebra generated by $\mathcal{C}$. For each $A \in \mathcal{A}$, set*

$$\bar{\mu}(A) = \sum_{i=1}^{k} \mu(B_i),$$

if the set A has the representation $A = \bigcup_{i=1}^{k} B_i$ for some $B_1, \ldots, B_k \in \mathcal{C}, k < \infty$ with $B_i \cap B_j = \emptyset$ for $i \neq j$. Then,

(i) $\bar{\mu}$ is independent of the representation of A as $A = \bigcup_{i=1}^{k} B_i$;

(ii) $\bar{\mu}$ is finitely additive on $\mathcal{A}$, i.e., $A, B \in \mathcal{A}, A \cap B = \emptyset \Rightarrow \bar{\mu}(A \cup B) = \bar{\mu}(A) + \bar{\mu}(B)$; and

(iii) $\bar{\mu}$ is countably additive on $\mathcal{A}$, i.e., if $A_n \in \mathcal{A}$ for all $n \geq 1$, $A_i \cap A_j = \emptyset$ for all $i = j$, and $\bigcup_{n\geq 1} A_n \in \mathcal{A}$, then

$$\bar{\mu}\Big(\bigcup_{n\geq 1} A_n\Big) = \sum_{n=1}^{\infty} \bar{\mu}(A_n).$$

Proof: Parts (i) and (ii) are easy to verify. Turning to part (iii), let each $n \geq 1$, $A_n = \bigcup_{j=1}^{k_n} B_{nj}$, $B_{nj} \in \mathcal{C}$, $\{B_{nj}\}_{j=1}^{k_n}$ disjoint. Since $\bigcup_{n\geq 1} A_n \in \mathcal{A}$ then exist disjoint sets $\{B_i\}_{i=1}^{k} \subset \mathcal{C}$ such that $\bigcup_{n\geq 1} A_n = \bigcup_{i=1}^{k} B_i$. Now

$$\begin{aligned} B_i &= B_i \cap \Big(\bigcup_{n\geq 1} A_n\Big) = \bigcup_{n\geq 1} (B_i \cap A_n) \\ &= \bigcup_{n\geq 1} \bigcup_{j=1}^{k_n} (B_i \cap B_{nj}). \end{aligned}$$

Since for all i, $B_i \in \mathcal{C}$, $B_i \cap B_{nj} \in \mathcal{C}$ for all j, n and μ is a measure on $\mathcal{C}$

$$\mu(B_i) = \sum_{n\geq 1} \sum_{j=1}^{k_n} \mu(B_i \cap B_{nj}).$$

Thus,

$$\bar{\mu}\Big(\bigcup_{n\geq 1} A_n\Big) = \sum_{i=1}^{k} \mu(B_i) = \sum_{i=1}^{k} \sum_{n\geq 1} \Big(\sum_{j=1}^{k_n} \mu(B_i \cap B_{nj})\Big)$$

$$
\begin{aligned}
&= \sum_{n \geq 1} \bigg(\sum_{i=1}^{k} \sum_{j=1}^{k_n} \mu(B_i \cap B_{nj}) \bigg) \\
&= \sum_{n \geq 1} \bar{\mu}(A_n),
\end{aligned}
$$

since

$$
\begin{aligned}
A_n &= A_n \cap \bigcup_{i=1}^{k} B_i \\
&= \bigcup_{i=1}^{k} \bigcup_{j=1}^{k_n} (B_i \cap B_{nj}).
\end{aligned}
$$

□

Thus, the set function $\bar{\mu}$ defined above is a measure on $\mathcal{A}$. To go beyond $\mathcal{A} = \mathcal{A}(\mathcal{C})$ to $\sigma\langle\mathcal{A}\rangle$, which is also the same as $\sigma\langle\mathcal{C}\rangle$ (see Problem 1.19 (ii)), the first step involves using the given set function μ on $\mathcal{C}$ to define a set function μ^* on the class of all subsets of Ω, i.e., on $\mathcal{P}(\Omega)$, where for any $A \in \mathcal{P}(\Omega)$, $\mu^*(A)$ is an estimate 'from above' of the μ-content of A.

Definition 1.3.3: Given a measure μ on a semialgebra $\mathcal{C}$, the *outer measure induced by* μ is the set function μ^*, defined on $\mathcal{P}(\Omega)$, as

$$
\mu^*(A) \equiv \inf \bigg\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C},\ A \subset \bigcup_{n \geq 1} A_n \bigg\}. \tag{3.1}
$$

Thus, a given set A is covered by countable unions of sets from $\mathcal{C}$ and the sums of the measures on such covers are computed and $\mu^*(A)$ is the greatest lower bound one can get in this way. It is not difficult to show that on $\mathcal{C}$ and $\mathcal{A}$, this is not an overestimate. That is, $\mu^* = \mu$ on $\mathcal{C}$ and on $\mathcal{A}$, $\mu^* = \bar{\mu}$. Now suppose $\mu(\Omega) < \infty$. Let $A \subset \Omega$ be a set such that the overestimates of the μ-contents of A and A^c, namely, $\mu^*(A)$ and $\mu^*(A^c)$, add up to $\mu(\Omega)$, i.e.,

$$
\mu(\Omega) = \mu^*(\Omega) = \mu^*(A) + \mu^*(A^c). \tag{3.2}
$$

Then, there is no room for error and the estimates $\mu^*(A)$ and $\mu^*(A^c)$ are in fact not overestimates at all. In this case, the set A may be considered 'exactly measurable' by this approximation procedure. This observation motivates the following definition.

Definition 1.3.4: A set A is said to be μ^*-*measurable* if

$$
\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset \Omega. \tag{3.3}
$$

In other words, an analog of (3.2) should hold in every portion E of Ω for a set A to be μ^*-measurable.

It can be shown (Problem 1.20) that μ^* defined in (3.1) satisfies:

$$\mu^*(\emptyset) = 0, \tag{3.4}$$
$$A \subset B \Rightarrow \mu^*(A) \le \mu^*(B), \tag{3.5}$$

and for any $\{A_n\}_{n\ge1} \subset \mathcal{P}(\Omega)$,

$$\mu^*\Big(\bigcup_{n\ge1} A_n\Big) \le \sum_{n=1}^{\infty} \mu^*(A_n)\,. \tag{3.6}$$

Definition 1.3.5: Any set function $\mu^* : \mathcal{P}(\Omega) \to \bar{\mathbb{R}}_+ \equiv [0,\infty]$ satisfying (3.4)–(3.6) is called an *outer measure on* Ω.

The following result (due to C. Caratheodory) yields a measure space on Ω starting from a general outer measure μ^* that need not arise from a measure μ as in (3.1).

Theorem 1.3.2: *Let μ^* be an outer measure on Ω, i.e., it satisfies (3.4)–(3.6). Let $\mathcal{M} \equiv \mathcal{M}_{\mu^*} \equiv \{A : A$ is μ^*-measurable, i.e., A satisfies (3.3)$\}$. Then*

(i) $\mathcal{M}$ is a σ-algebra,

(ii) μ^* restricted to $\mathcal{M}$ is a measure, and

(iii) $\mu^*(A) = 0 \Rightarrow \mathcal{P}(A) \subset \mathcal{M}$.

Proof: From (3.3), it follows that $\emptyset \in \mathcal{M}$ and that $A \in \mathcal{M} \Rightarrow A^c \in \mathcal{M}$. Next, it will be shown that $\mathcal{M}$ is closed under finite unions. Let $A_1, A_2 \in \mathcal{M}$. Then, for any $E \subset \Omega$,

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A_1) + \mu^*(E \cap A_1^c) \quad (\text{since } A_1 \in \mathcal{M}) \\ &= \mu^*(E \cap A_1 \cap A_2) + \mu^*(E \cap A_1 \cap A_2^c) \\ &\quad + \mu^*(E \cap A_1^c \cap A_2) + \mu^*(E \cap A_1^c \cap A_2^c) \quad (\text{since } A_2 \in \mathcal{M}). \end{aligned}$$

But $(A_1 \cap A_2) \cup (A_1 \cap A_2^c) \cup (A_1^c \cap A_2) = A_1 \cup A_2$. Since μ^* is subadditive, it follows that

$$\mu^*(E \cap (A_1 \cup A_2)) \le \mu^*(E \cap A_1 \cap A_2) + \mu^*(E \cap A_1 \cap A_2^c) + \mu^*(E \cap A_1^c \cap A_2).$$

Thus

$$\mu^*(E) \ge \mu^*(E \cap (A_1 \cup A_2)) + \mu^*(E \cap (A_1 \cup A_2)^c).$$

The subadditivity of μ^* yields the opposite inequality and so, $A_1 \cup A_2 \in \mathcal{M}$ and hence, $\mathcal{M}$ is an algebra. To show that $\mathcal{M}$ is a σ-algebra, it suffices to show that $\mathcal{M}$ is closed under monotone unions, i.e., $A_n \in \mathcal{M}, A_n \subset A_{n+1}$ for all $n \ge 1 \Rightarrow A \equiv \bigcup_{n\ge1} A_n \in \mathcal{M}$. Let $B_1 = A_1$ and $B_n = A_n \cap A_{n-1}^c$

for all $n \geq 2$. Then, for all $n \geq 1$, $B_n \in \mathcal{M}$ (since $\mathcal{M}$ is an algebra), $\bigcup_{j=1}^n B_j = A_n$, and $\bigcup_{j=1}^\infty B_j = A$. Hence, for any $E \subset \Omega$,

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap A_n) + \mu^*(E \cap A_n^c) \\
&= \mu^*(E \cap A_n \cap B_n) + \mu^*(E \cap A_n \cap B_n^c) + \mu^*(E \cap A_n^c) \\
&\qquad\qquad\qquad\qquad\qquad\qquad \text{(since } B_n \in \mathcal{M}\text{)} \\
&= \mu^*(E \cap B_n) + \mu^*(E \cap A_{n-1}) + \mu^*(E \cap A_n^c) \\
&= \sum_{j=1}^n \mu^*(E \cap B_j) + \mu^*(E \cap A_n^c) \quad \text{(by iteration)} \\
&\geq \sum_{j=1}^n \mu^*(E \cap B_j) + \mu^*(E \cap A^c) \quad \text{(by monotonicity).}
\end{aligned}$$

Now letting $n \to \infty$, and using the subadditivity of μ^* and the fact that $\bigcup_{j=1}^\infty B_j = A$, one gets

$$\begin{aligned}
\mu^*(E) &\geq \sum_{j=1}^\infty \mu^*(E \cap B_j) + \mu^*(E \cap A^c) \\
&\geq \mu^*(E \cap A) + \mu^*(E \cap A^c).
\end{aligned}$$

This completes the proof of part (i).

To prove part (ii), let $\{B_n\}_{n \geq 1} \subset \mathcal{M}$ and $B_i \cap B_j = \emptyset$ for $i \neq j$. Let $A_j = \bigcup_{i=j}^\infty B_i$, $j \in \mathbb{N}$. Then, by (i), $A_j \in \mathcal{M}$ for all $j \in \mathbb{N}$ and so

$$\begin{aligned}
\mu^*(A_1) &= \mu^*(A_1 \cap B_1) + \mu^*(A_1 \cap B_1^c) \quad \text{(since } B_1 \in \mathcal{M}\text{)} \\
&= \mu^*(B_1) + \mu^*(A_2) \\
&= \mu^*(B_1) + \mu^*(B_2) + \mu^*(A_3) \quad \text{(by iteration)} \\
&= \sum_{i=1}^n \mu^*(B_i) + \mu^*(A_{n+1}) \quad \text{(by iteration)} \\
&\geq \sum_{i=1}^n \mu^*(B_i) \quad \text{for all} \quad n \geq 1.
\end{aligned}$$

Now letting $n \to \infty$, one has $\mu^*(A_1) \geq \sum_{i=1}^\infty \mu^*(B_i)$. By subadditivity of μ^*, the opposite inequality holds and so

$$\mu^*(A_1) = \mu^*\left(\bigcup_{i=1}^\infty B_i\right) = \sum_{i=1}^\infty \mu^*(B_i),$$

proving (ii).

As for (iii), note that by monotonicity, $\mu^*(A) = 0$, $B \subset A \Rightarrow \mu^*(B) = 0$ and hence, for any E, $\mu^*(E \cap B) = 0$. Since $\mu^*(E) \geq \mu^*(E \cap B^c)$, this implies $\mu^*(E) \geq \mu^*(E \cap B^c) + \mu^*(E \cap B)$. The opposite inequality holds by the subadditivity of μ^*. So $B \in \mathcal{M}$, and (iii) is proved. $\square$

Definition 1.3.6: A measure space $(\Omega, \mathcal{F}, \nu)$ is called *complete* if for any $A \in \mathcal{F}$ with $\nu(A) = 0 \Rightarrow \mathcal{P}(A) \subset \mathcal{F}$.

Thus, by part (iii) of the above theorem, $(\Omega, \mathcal{M}_{\mu^*}, \mu^*)$ is a *complete measure space*. Now the above theorem is applied to a μ^* that is generated by a given measure μ on a semialgebra $\mathcal{C}$ via (3.1).

Theorem 1.3.3: (*Caratheodory's extension theorem*). *Let μ be a measure on a semialgebra $\mathcal{C}$ and let μ^* be the set function induced by μ as defined by (3.1). Then,*

(i) μ^* is an outer measure,

(ii) $\mathcal{C} \subset \mathcal{M}_{\mu^*}$, and

(iii) $\mu^* = \mu$ on $\mathcal{C}$, where $\mathcal{M}_{\mu^*}$ is as in Theorem 1.3.2.

Proof: The proof of (i) involves verifying (3.4)–(3.6), which is left as an exercise (Problem 1.20). To prove (ii), let $A \in \mathcal{C}$. Let $E \subset \Omega$ and $\{A_n\}_{n\geq 1} \subset \mathcal{C}$ be such that $E \subset \bigcup_{n\geq 1} A_n$. Then, for all $i \in \mathbb{N}$, $A_i = (A_i \cap A) \cup (A_i \cap B_1) \cup \ldots \cup (A_i \cap B_k)$ where $B_1, \ldots, B_k$ are disjoint sets in $\mathcal{C}$ such that $\bigcup_{j=1}^k B_j = A^c$. Since μ is finitely additive on $\mathcal{C}$,

$$\begin{aligned} \mu(A_i) &= \mu(A_i \cap A) + \sum_{j=1}^k \mu(A_i \cap B_j) \\ \Rightarrow \quad \sum_{n=1}^\infty \mu(A_n) &= \sum_{n=1}^\infty \mu(A_n \cap A) + \sum_{n=1}^\infty \sum_{j=1}^k \mu(A_n \cap B_j) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c), \end{aligned}$$

since $\{A_n \cap A\}_{n\geq 1}$ and $\{A_n \cap B_j : 1 \leq j \leq k, n \geq 1\}$ are both countable subcollections of $\mathcal{C}$ whose unions cover $E \cap A$ and $E \cap A^c$, respectively. From the definition of $\mu^*(E)$, it now follows that

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Now the subadditivity of μ^* completes the proof of part (ii).

To prove (iii), let $A \in \mathcal{C}$. Then, by definition, $\mu^*(A) \leq \mu(A)$. If $\mu^*(A) = \infty$, then $\mu(A) = \infty = \mu^*(A)$. If $\mu^*(A) < \infty$, then by the definition of 'infimum,' for any $\epsilon > 0$, there exists $\{A_n\}_{n\geq 1} \subset \mathcal{C}$ such that $A \subset \bigcup_{n\geq 1} A_n$ and

$$\mu^*(A) \leq \sum_{n=1}^\infty \mu(A_n) \leq \mu^*(A) + \epsilon.$$

But $A = A \cap (\bigcup_{n \geq 1} A_n) = \bigcup_{n \geq 1} (A \cap A_n)$. Note that the set function $\bar{\mu}$ defined in Proposition 1.3.1 is a measure on $\mathcal{A}(\mathcal{C})$ and it coincides with μ on $\mathcal{C}$. Since $A, A \cap A_n \in \mathcal{C}$ for all $n \geq 1$, by Proposition 1.2.3 (b) applied to $\bar{\mu}$,

$$\mu(A) = \bar{\mu}(A) \leq \sum_{n=1}^{\infty} \bar{\mu}(A \cap A_n) \leq \sum_{n=1}^{\infty} \bar{\mu}(A_n) = \sum_{n=1}^{\infty} \mu(A_n) \leq \mu^*(A) + \epsilon.$$

(Alternately, w.l.o.g., assume that $\{A_n\}_{n \geq 1}$ are disjoint. Then $A = A \cap \bigcup_{n \geq 1} A_n = \bigcup_{n \geq 1} (A \cap A_n)$. Since $A \cap A_n \in \mathcal{C}$ for all n, by the countable additivity of μ on $\mathcal{C}$, $\mu(A) = \sum_{n=1}^{\infty} \mu(A \cap A_n) \leq \sum_{n=1}^{\infty} \mu(A_n) = \mu^*(A) + \epsilon$.) Since $\epsilon > 0$ is arbitrary, this yields $\mu(A) \leq \mu^*(A)$. □

Thus, given a measure μ on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$, there is a complete measure space $(\Omega, \mathcal{M}_{\mu^*}, \mu^*)$ such that $\mathcal{M}_{\mu^*} \supset \mathcal{C}$ and μ^* restricted to $\mathcal{C}$ equals μ. For this reason, μ^* is called an *extension* of μ. The measure space $(\Omega, \mathcal{M}_{\mu^*}, \mu^*)$ is called the *Caratheodory extension* of μ. Since $\mathcal{M}_{\mu^*}$ is a σ-algebra and contains $\mathcal{C}$, $\mathcal{M}_{\mu^*}$ must contain $\sigma\langle\mathcal{C}\rangle$, the σ-algebra generated by $\mathcal{C}$, and thus, $(\Omega, \sigma\langle\mathcal{C}\rangle, \mu^*)$ is also a measure space. However, the latter may *not* be *complete* (see Section 1.4).

Now the above method is applied to the construction of Lebesgue-Stieltjes measures on $\mathbb{R}$ and $\mathbb{R}^2$.

1.3.2 Lebesgue-Stieltjes measures on $\mathbb{R}$

Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing. For $x \in \mathbb{R}$, let $F(x+) \equiv \lim_{y \downarrow x} F(y)$, and $F(x-) \equiv \lim_{y \uparrow x} F(y)$. Set $F(\infty) = \lim_{x \uparrow \infty} F(x)$ and $F(-\infty) = \lim_{x \downarrow -\infty} F(y)$. Let

$$\mathcal{C} \equiv \Big\{(a, b] : -\infty \leq a \leq b < \infty\Big\} \cup \Big\{(a, \infty) : -\infty \leq a < \infty\Big\}. \qquad (3.7)$$

Define

$$\begin{aligned} \mu_F((a, b]) &= F(b+) - F(a+), \\ \mu_F((a, \infty)) &= F(\infty) - F(a+). \end{aligned} \qquad (3.8)$$

Then, it may be verified that

(i) $\mathcal{C}$ is a semialgebra;

(ii) μ_F is a measure on $\mathcal{C}$. (For (ii), one needs to use the Heine-Borel theorem. See Problems 1.22 and 1.23.)

Let $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ be the Caratheodory extension of μ_F, i.e., the measure space constructed as in the above two theorems.

Definition 1.3.7: Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing. The (measure) space $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ is called a *Lebesgue-Stieltjes measure space* and μ_F^* is the *Lebesgue-Stieltjes measure* generated by F.

Since $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$, the class of all Borel sets of $\mathbb{R}$, every Lebesgue-Stieltjes measure μ_F^* is also a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Note also that μ_F^* is finite on bounded intervals.

Conversely, given any *Radon measure* μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e., a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that is finite on bounded intervals, set

$$F(x) = \begin{cases} \mu((0,x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((x,0]) & \text{if } x \le 0. \end{cases}$$

Then $\mu_F = \mu$ on $\mathcal{C}$. By the uniqueness of the extension (discussed later in this section, see also Theorem 1.2.4), it follows that μ_F^* coincides with μ on $\mathcal{B}(\mathbb{R})$. Thus, every Radon measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is necessarily a Lebesgue-Stieltjes measure.

Definition 1.3.8: (*Lebesgue Measure on* $\mathbb{R}$). When $F(x) \equiv x, x \in \mathbb{R}$, the measure μ_F^* is called the *Lebesgue measure* and the σ-algebra $\mathcal{M}_{\mu_F^*}$ is called the class of *Lebesgue measurable sets*.

The Lebesgue measure will be denoted by $m(\cdot)$ or $\mu_L(\cdot)$. Given below are some important results on $m(\cdot)$.

(i) It follows from equation (3.1), that $\mu_F^*(\{x\}) = F(x+) - F(x-)$ and hence $= 0$ if F is continuous at x. Thus $m(\{x\}) \equiv 0$ on $\mathbb{R}$.

(ii) By countable additivity of $m(\cdot)$, $m(A) = 0$ for any countable set A.

(iii) (*Cantor set*). There exists an uncountable set C such that $m(C) = 0$. An example is the *Cantor set* constructed as follows: Start with $I_0 = [0,1]$. Delete the open middle third, i.e., $\left(\frac{1}{3}, \frac{2}{3}\right)$. Next from the closed intervals $I_{11} = [0, \frac{1}{3}]$ and $I_{12} = [\frac{2}{3}, 1]$ delete the open middle thirds, i.e., $\left(\frac{1}{9}, \frac{2}{9}\right)$ and $\left(\frac{7}{9}, \frac{2}{9}\right)$, respectively. Repeat this process of deleting the middle third from each of the remaining closed intervals. Thus at stage n there will be 2^{n-1} new closed intervals and 2^{n-1} deleted open intervals, each of length $\frac{1}{3^n}$. Let U_n denote the union of all the deleted open intervals at the nth stage. Then $\{U_n\}_{n\ge1}$ are disjoint open sets. Let $U \equiv \bigcup_{n\ge1} U_n$. By countable additivity

$$m(U) = \sum_{n=1}^{\infty} m(U_n) = \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = 1.$$

Let $C \equiv [0,1] - U$. Since U is open and [0,1] is closed, C is nonempty. It can be shown that $C \equiv \{x : x = \sum_1^\infty \frac{a_i}{3^i}, a_i = 0 \text{ or } 2\}$ (Problem

1.33). Thus C can be mapped in (1–1) manner on to the set of all sequences $\{\delta_i\}_{i\geq 1}$ such that $\delta_i = 0$ or 2. But this set is uncountable. Since $m([0,1]) = 1$, it follows that $m(C) = 0$. For more properties of the Cantor set, see Rudin (1976) and Chapter 4.

(iv) $m(\cdot)$ is invariant under reflection and translation. That is, for any E in $\mathcal{B}(\mathbb{R})$,

$$m(-E) = m(E) \quad \text{and} \quad m(E+c) = m(E)$$

for all c in $\mathbb{R}$ where $-E = \{-x : x \in E\}$ and $E + c = \{y : y = x + c, x \in E\}$. This follows from Theorem 1.2.4 and the fact that the claim holds for intervals (cf. Problem 2.48).

(v) There exists a subset $A \subset \mathbb{R}$ such that $A \notin \mathcal{M}_m$. That is, there exists a *non-Lebesgue measurable set*. The proof of this requires the use of the axiom of choice (cf. A.1). For a proof see Royden (1988).

1.3.3 Lebesgue-Stieltjes measures on $\mathbb{R}^2$

Let $F : \mathbb{R}^2 \to \mathbb{R}$ satisfy

$$F(a_2, b_2) - F(a_2, b_1) - F(a_1, b_2) + F(a_1, b_1) \geq 0, \tag{3.9}$$

and

$$F(a_2, b_2) - F(a_1, b_1) \geq 0, \tag{3.10}$$

for all $a_1 \leq a_2$, $b_1 \leq b_2$. Extend F to $\bar{\mathbb{R}}^2$ by appropriate limiting procedure. Let

$$\mathcal{C}_2 \equiv \{I_1 \times I_2 : I_1, I_2 \in \mathcal{C}\}, \tag{3.11}$$

where $\mathcal{C}$ is as in (3.7). Next, for $I_1 = (a_1, a_2]$, $I_2 = (b_1, b_2]$, $-\infty < a_1, a_2, b_1, b_2 < \infty$, set

$$\mu_F(I_1 \times I_2) \equiv F(a_2+, b_2+) - F(a_2+, b_1+) - (F(a_1+, b_2+) + F(a_1+, b_1+)), \tag{3.12}$$

where for any $a, b \in \mathbb{R}$, $F(a+, b+)$ is defined as

$$F(a+, b+) \equiv \lim_{a' \downarrow a, b' \downarrow b} F(a', b').$$

Note that by (3.10), the limit exists and hence, $F(a+, b+)$ is well defined. Further, by (3.9), the right side of (3.12) is nonnegative. Next extend the definition of μ_F to unbounded sets in $\mathcal{C}_2$ by the limiting procedure:

$$\mu_F(I_1 \times I_2) = \lim_{n\to\infty} \mu_F\big((I_1 \times I_2) \cap J_n\big), \tag{3.13}$$

where $J_n = (-n, n] \times (-n, n]$. Then it may be verified (Problems 1.24, 1.25) that

(i) $\mathcal{C}_2$ is a semialgebra

(ii) μ_F is a measure on $\mathcal{C}_2$.

Let $(\mathbb{R}^2, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ be the measure space constructed as in the above two theorems. The measure μ_F^* is called the *Lebesgue-Stieltjes measure* generated by F and $\mathcal{M}_{\mu_F^*}$ a *Lebesgue-Stieltjes σ-algebra*. Again, in this case, $\mathcal{M}_{\mu^*}$ includes the σ-algebra $\sigma\langle\mathcal{C}\rangle \equiv \mathcal{B}(\mathbb{R}^2)$ and so $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \mu_F^*)$ is also a measure space. If $F(a,b) = ab$, then μ_F is called the *Lebesgue measure on* $\mathbb{R}^2$.

A similar construction holds for any $\mathbb{R}^k$, $k < \infty$.

1.3.4 More on extension of measures

Next the uniqueness of the extension μ^* and approximation of the μ^*-measure of a set in $\mathcal{M}_{\mu^*}$ by that of a set from the algebra $\mathcal{A} \equiv \mathcal{A}(\mathcal{C})$ are considered. As in the case of measures defined on an algebra, a measure μ on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$ is said to be *σ-finite* if there exists a *countable* collection $\{A_n\}_{n\geq 1} \subset \mathcal{C}$ such that (i) $\mu(A_n) < \infty$ for each $n \geq 1$ and (ii) $\bigcup_{n\geq 1} A_n = \Omega$. The following approximation result holds.

Theorem 1.3.4: *Let $A \in \mathcal{M}_{\mu^*}$ and $\mu^*(A) < \infty$. Then, for each $\epsilon > 0$, there exist $B_1, B_2, \ldots, B_k \in \mathcal{C}$, $k < \infty$ with $B_i \cap B_j = \emptyset$ for $1 \leq i \neq j \leq k$, such that*

$$\mu^*\Big(A \bigtriangleup \bigcup_{j=1}^{k} B_j\Big) < \epsilon,$$

where for any two sets E_1 and E_2, $E_1 \bigtriangleup E_2$ is the symmetric difference of E_1 and E_2, defined by $E_1 \bigtriangleup E_2 \equiv (E_1 \cap E_2^c) \cup (E_1^c \cap E_2)$.

Proof: By definition of μ^*, $\mu^*(A) < \infty$ implies that for every $\epsilon > 0$, there exist $\{B_n\}_{n\geq 1} \subset \mathcal{C}$ such that $A \subset \bigcup_{n\geq 1} B_n$ and

$$\mu^*(A) \leq \sum_{n=1}^{\infty} \mu(B_n) \leq \mu^*(A) + \epsilon/2 < \infty.$$

Since $B_n \in \mathcal{C}$, B_n^c is a finite union of disjoint sets from $\mathcal{C}$. W.l.o.g., it can be assumed that $\{B_n\}_{n\geq 1}$ are disjoint. (Otherwise, one can consider the sequence $B_1, B_2 \cap B_1^c, B_3 \cap B_2^c \cap B_1^c, \ldots$.) Next, $\sum_{n=1}^{\infty} \mu(B_n) < \infty$ implies that for every $\epsilon > 0$, there exists $k \in \mathbb{N}$ such that $\sum_{n=k+1}^{\infty} \mu(B_n) < \frac{\epsilon}{2}$. Since both A and $\bigcup_{j=1}^{k} B_j$ are subsets of $\bigcup_{n\geq 1} B_n$,

$$\mu^*\Big(A \bigtriangleup \Big[\bigcup_{j=1}^{k} B_j\Big]\Big) \leq \mu^*\Big(A^c \cap \Big[\bigcup_{j=1}^{\infty} B_j\Big]\Big) + \mu^*\Big(\bigcup_{j=k+1}^{\infty} B_j\Big).$$

But since μ^* is a measure on $(\Omega, \mathcal{M}_{\mu^*})$, $\mu^*(\bigcup_{n\geq 1} B_n) = \mu^*(A) + \mu^*((\bigcup_{n\geq 1} B_n) \cap A^c)$. Further, since $\mu^*(A) < \infty$,

$$\begin{aligned}
& \mu^*\Big(\Big[\bigcup_{n\geq 1} B_n\Big] \cap A^c\Big) \\
= \ & \mu^*\Big(\bigcup_{n\geq 1} B_n\Big) - \mu^*(A)) \\
\leq \ & \sum_{n=1}^{\infty} \mu^*(B_n) - \mu^*(A), \quad \text{(since } \mu^* \text{ is countably subadditive)} \\
= \ & \sum_{n=1}^{\infty} \mu(B_n) - \mu^*(A), \quad \text{(since } \mu^* = \mu \text{ on } \mathcal{C}) \\
< \ & \frac{\epsilon}{2} \quad \text{(by the choice of } \{B_n\}_{n\geq 1}).
\end{aligned}$$

Also, by the definition of k,

$$\mu^*\Big(\bigcup_{j=k+1}^{\infty} B_j\Big) \leq \sum_{j=k+1}^{\infty} \mu^*(B_j) = \sum_{j=k+1}^{\infty} \mu(B_j) < \frac{\epsilon}{2}.$$

Thus, $\mu^*(A \triangle [\bigcup_{j=1}^{k} B_j]) < \epsilon$. This completes the proof of the theorem. $\Box$

Thus, every μ^*-measurable set of finite measure is nearly a finite union of disjoint elements from the semialgebra $\mathcal{C}$. This was enunciated by J.E. Littlewood as the first of his three principles of approximation (the other two being: every Lebesgue measurable function is nearly continuous (cf. Theorem 2.5.12) and every almost everywhere convergent sequence of functions on a finite measure space is nearly uniformly convergent (Egorov's theorem) cf. Theorem 2.5.11).

One may strengthen Theorem 1.3.4 to prove the following result on regularity of *Radon* measures on $\big(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k)\big)$ (cf. Problem 1.32). See also Rudin (1987), Chapter 2.

Corollary 1.3.5: (*Regularity of measures*). *Let μ be a Radon measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ for some $k \in \mathbb{N}$, i.e., $\mu(A) < \infty$ for all bounded sets $A \in \mathcal{B}(\mathbb{R}^k)$. Let $A \in \mathcal{B}(\mathbb{R}^k)$ be such that $\mu(A) < \infty$. Then, for each $\epsilon > 0$, there exist a compact set K and an open set G such that $K \subset A \subset G$ and $\mu(G \setminus K) < \epsilon$.*

The following uniqueness result can be established by an application of the above approximation theorem (Theorem 1.3.4) or by applying the π-λ theorem (Corollary 1.1.3), as in Theorem 1.2.4 (Problem 1.26).

Theorem 1.3.6: (*Uniqueness*). *Let μ be a σ-finite measure on a semialgebra $\mathcal{C}$. Let ν be a measure on the measurable space $(\Omega, \sigma\langle\mathcal{C}\rangle)$ such that $\nu = \mu$ on $\mathcal{C}$. Then $\nu = \mu^*$ on $\sigma\langle\mathcal{C}\rangle$.*

An application of Theorem 1.3.4 yields an useful approximation result known as Lusin's theorem (see Theorem 2.1.3 in Section 2.1) for approximating Borel measurable functions by continuous functions.

1.4 Completeness of measures

Recall from Definition 1.3.6 that a measure space $(\Omega, \mathcal{F}, \mu)$ is called *complete* if for any $A \in \mathcal{F}$, $\mu(A) = 0$, $B \subset A \ \Rightarrow B \in \mathcal{F}$. That is, for any set A in $\mathcal{F}$ whose μ measure is zero, all subsets of a set A are also in $\mathcal{F}$. For example, the very construction of the Lebesgue-Stieltjes measure for a nondecreasing F on $\mathbb{R}$, discussed in Section 1.3, yields a complete measure space $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$. The Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is a sub-σ-algebra of $\mathcal{M}_{\mu_F^*}$ and $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F^*)$ need not be complete. For example, if μ_F is the Lebesgue measure, then the Cantor set C (Section 1.3.2) has measure 0 and hence $\mathcal{M} \equiv$ the Lebesgue σ-algebra contains the power set of C and hence has cardinality larger than that of $\mathbb{R}$. It can be shown that the cardinality of $\mathcal{B}(\mathbb{R})$ is the same as that of $\mathbb{R}$ (see Hewitt and Stromberg (1965)). For another example, if F is a degenerate distribution at 0, i.e., $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x \geq 0$, then $\mathcal{M}_{\mu_F^*} = \mathcal{P}(\mathbb{R})$, the power set of $\mathbb{R}$ (Problem 1.28), and hence $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$ is not complete. The same is true for any discrete distribution function. However, it is always possible to complete an incomplete measure space $(\Omega, \mathcal{F}, \mu)$ by adding new sets to $\mathcal{F}$. This procedure is discussed below.

Theorem 1.4.1: *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Let $\tilde{\mathcal{F}} \equiv \{A : B_1 \subset A \subset B_2$ for some $B_1, B_2 \in \mathcal{F}$ satisfying $\mu(B_2 \setminus B_1) = 0\}$. For any $A \in \tilde{\mathcal{F}}$, set $\tilde{\mu}(A) = \mu(B_1) = \mu(B_2)$ for any pair of sets $B_1, B_2 \in \mathcal{F}$ with $B_1 \subset A \subset B_2$ and $\mu(B_2 \setminus B_1) = 0$. Then*

(i) *$\tilde{\mathcal{F}}$ is a σ-algebra and $\mathcal{F} \subset \tilde{\mathcal{F}}$,*

(ii) *$\tilde{\mu}$ is well defined,*

(iii) *$(\Omega, \tilde{\mathcal{F}}, \tilde{\mu})$ is a complete measure space and $\tilde{\mu} = \mu$ on $\mathcal{F}$.*

Proof:

(i) Since $A \in \tilde{\mathcal{F}}$, there exist $B_1, B_2 \in \mathcal{F}$, $B_1 \subset A \subset B_2$ and $\mu(B_2 \setminus B_1) = 0$. Clearly, $B_2^c \subset A^c \subset B_1^c$, and $B_1^c, B_2^c \in \mathcal{F}$ and $\mu(B_1^c \setminus B_2^c) = \mu(B_2 \setminus B_1) = 0$ and so $A^c \in \tilde{\mathcal{F}}$. Next, let $\{A_n\}_{n=1}^{\infty} \subset \tilde{\mathcal{F}}$ and $A = \bigcup_{n \geq 1} A_n$. Then, for each n there exist B_{1n} and B_{2n} in $\mathcal{F}$ such that $B_{1n} \subset A_n \subset B_{2n}$ and $\mu(B_{2n} \setminus B_{1n}) = 0$. Let $B_1 = \bigcup_{n \geq 1} B_{1n}$ and $B_2 = \bigcup_{n \geq 1} B_{2n}$. Then $B_1 \subset A \subset B_2$, $B_1, B_2 \in \mathcal{F}$ and $B_2 \setminus B_1 \subset \bigcup_{n \geq 1}(B_{2n} \setminus B_{1n})$ and hence $\mu(B_2 \setminus B_1) \leq \sum_{n=1}^{\infty} \mu(B_{2n} \setminus B_{1n}) = 0$. Thus, $A \in \tilde{\mathcal{F}}$ and hence $\tilde{\mathcal{F}}$ is a σ-algebra. Clearly, $\mathcal{F} \subset \tilde{\mathcal{F}}$ since for every $A \in \mathcal{F}$, one may take $B_1 = B_2 = A$.

(ii) Let $B_1 \subset A \subset B_2$, $B_1' \subset A \subset B_2'$, $B_1, B_1', B_2, B_2' \in \mathcal{F}$ and $\mu(B_2 \setminus B_1) = 0 = \mu(B_2' \setminus B_1')$. Then $B_1 \cup B_1' \subset A \subset B_2 \cap B_2'$ and $(B_2 \cap B_2') \setminus (B_1 \cup B_1') = (B_2 \cap B_2') \cap (B_1^c \cap B_1'^c) \subset B_2 \cap B_1^c$. Thus

$$\mu([B_2 \cap B_2'] \setminus [B_1 \cup B_1']) = 0.$$

Hence, $\mu(B_2) = \mu(B_1) + \mu(B_2 \setminus B_1) = \mu(B_1) \leq \mu(B_1 \cup B_1') = \mu(B_2 \cap B_2') \leq \mu(B_2')$. By symmetry $\mu(B_2') \leq \mu(B_2)$ and so $\mu(B_2) = \mu(B_2')$. But $\mu(B_2) = \mu(B_1)$ and $\mu(B_2') = \mu(B_1')$ and also all four quantities agree.

(iii) It remains to show that $\tilde{\mu}$ is countably additive and complete on $\tilde{\mathcal{F}}$. Let $\{A_n\}_{n=1}^{\infty}$ be a disjoint sequence of sets from $\tilde{\mathcal{F}}$ and let $A = \bigcup_{n \geq 1} A_n$. Let $\{B_{1n}\}_{n \geq 1}, \{B_{2n}\}_{n \geq 1}, B_1, B_2$ be as in the proof of (i). Then, the fact that $\{A_n\}_{n=1}^{\infty}$ are disjoint implies $\{B_{1n}\}_{n=1}^{\infty}$ are also disjoint. And since $B_1 = \bigcup_{n \geq 1} B_{1n}$ and μ is a measure on $(\Omega, \mathcal{F})$,

$$\mu(B_1) \equiv \sum_{n=1}^{\infty} (B_{1n}).$$

Also, by definition of B_{1n}'s, $\mu(B_{1n}) = \tilde{\mu}(A_n)$ for all $n \geq 1$, and by (i), $\tilde{\mu}(A) = \mu(B_1)$. Thus,

$$\tilde{\mu}(A) = \mu(B_1) = \sum_{n=1}^{\infty} (B_{1n}) = \sum_{n=1}^{\infty} \tilde{\mu}(A_n),$$

establishing the countable additivity of $\tilde{\mu}$.

Next, suppose that $A \in \tilde{\mathcal{F}}$ and $\tilde{\mu}(A) = 0$. Then there exist $B_1, B_2 \in \mathcal{F}$ such that $B_1 \subset A \subset B_2$ and $\mu(B_2 \setminus B_1) = 0$. Further, by definition of $\tilde{\mu}$, $\mu(B_2) = \tilde{\mu}(A) = 0$. If $D \subset A$, then $\emptyset \subset D \subset B_2$ and $\mu(B_2 \setminus \emptyset) = 0$. Therefore, $D \in \tilde{\mathcal{F}}$ and hence $(\Omega, \tilde{\mathcal{F}}, \tilde{\mu})$ is complete.

Finally, if $A \in \mathcal{F}$, then take $B_1 = B_2 = A$ and so, $\tilde{\mu}(A) = \mu(B_1) = \mu(A)$, and thus, $\tilde{\mu} = \mu$ on $\mathcal{F}$. Hence, the proof of the theorem is complete. $\square$

1.5 Problems

1.1 Let Ω be a nonempty set. Show that $\mathcal{F} \equiv \{\Omega, \emptyset\}$ and $\mathcal{G} = \mathcal{P}(\Omega) \equiv \{A : A \subset \Omega\}$ are both σ-algebras.

1.2 Let Ω be a finite set, i.e., the number of elements in Ω is finite. Let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be an algebra. Show that $\mathcal{F}$ is a σ-algebra.

1.3 Show that $\mathcal{F}_6$ in Example 1.1.4 is a σ-algebra.

1.4 Let $\{\mathcal{F}_\theta : \theta \in \Theta\}$ be a family of σ-algebras on Ω. Show that $\mathcal{G} \equiv \bigcap_{\theta\in\Theta} \mathcal{F}_\theta$ is also a σ-algebra.

1.5 Let $\Omega = \{1,2,3\}$, $\mathcal{F}_1 = \{\{1\},\{2,3\},\Omega,\emptyset\}$, $\mathcal{F}_2 = \{\{1,2\},\{3\},\Omega,\emptyset\}$. Verify that $\mathcal{F}_1$ and $\mathcal{F}_2$ are both algebras (in fact, σ-algebras) but $\mathcal{F}_1 \cup \mathcal{F}_2$ is not an algebra.

1.6 Let Ω be a nonempty set and let $\mathcal{A} \equiv \{A_i : i \in \mathbb{N}\}$ be a *partition* of Ω, i.e., $A_i \cap A_j = \emptyset$ for all $i \neq j$ and $\bigcup_{n\geq 1} A_n = \Omega$. Let $\mathcal{F} = \{\bigcup_{i\in J} A_i : J \subset \mathbb{N}\}$ where, for $J = \emptyset$, $\bigcup_{i\in J} A_i \equiv \emptyset$. Show that $\mathcal{F}$ is a σ-algebra.

1.7 Let Ω be a nonempty set and let $\mathcal{B} \equiv \{B_i : 1 \leq i \leq k < \infty\} \subset \mathcal{P}(\Omega)$, $\mathcal{B}$ not necessarily a partition. Find $\sigma\langle\mathcal{B}\rangle$.

(**Hint:** For each $\delta = (\delta_1, \delta_2, \ldots, \delta_k)$, $\delta_i \in \{0,1\}$ let $B_\delta = \bigcap_{i=1}^{k} B_i(\delta_i)$, where $B_i(0) = B_i^c$ and $B_i(1) = B_i$, $i \geq 1$. Show that $\sigma(\mathcal{B}) = \{E : E = \bigcup_{\delta\in J} B_\delta,\ J \subset \{1,2,\ldots,k\}\}$.)

1.8 Let $\Omega \equiv \{1,2,\ldots\} = \mathbb{N}$ and $A_i \equiv \{j : j \in \mathbb{N}, j \geq i\}, i \in \mathbb{N}$. Show that $\sigma\langle\mathcal{A}\rangle = \mathcal{P}(\Omega)$ where $\mathcal{A} = \{A_i : i \in \mathbb{N}\}$.

1.9 (a) Show that every open set $A \subset \mathbb{R}$ is a countable union of open intervals.

(**Hint:** Use the fact that the set $\mathcal{Q}$ of all rational numbers is dense in $\mathbb{R}$.)

(b) Extend the above to $\mathbb{R}^k$ for any $k \in \mathbb{N}$.

(c) Strengthen (a) to assert that the open intervals in (a) can be chosen to be disjoint.

1.10 Show that in Example 1.1.6, $\mathcal{O}_j \subset \sigma\langle\mathcal{O}_i\rangle$ for all $1 \leq i,j \leq 4$.

1.11 For $k \in \mathbb{N}$, let $\mathcal{O}_5 \equiv \{(a_1,b_1] \times \ldots \times (a_k,b_k] : -\infty < a_i < b_i < \infty, 1 \leq i \leq k\}$ and $\mathcal{O}_6 \equiv \{[a_1,b_1] \times \ldots \times [a_k,b_k] : -\infty < a_i < b_i < \infty, 1 \leq i \leq k\}$. Show that $\sigma\langle\mathcal{O}_5\rangle = \sigma\langle\mathcal{O}_6\rangle = \mathcal{B}(\mathbb{R}^k)$.

1.12 Let $\mathcal{O}_S \equiv \{\{x\} : x \in \mathbb{R}^k\}$ be the class of all singletons in $\mathbb{R}^k$, $k \in \mathbb{N}$. Show that $\sigma\langle\mathcal{O}_S\rangle$ is *properly* contained in $\mathcal{B}(\mathbb{R}^k)$.

(**Hint:** Show that $\sigma\langle\mathcal{O}_S\rangle$ coincides with $\mathcal{F}_6$ of Example 1.1.4).

1.13 Let $\bar{\mathbb{R}} \equiv \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$ be the extended real line. The Borel σ-algebra on $\bar{\mathbb{R}}$, denoted by $\mathcal{B}(\bar{\mathbb{R}})$, is defined as the σ-algebra on $\bar{\mathbb{R}}$ generated by the collection $\mathcal{B}(\mathbb{R}) \cup \{\infty\} \cup \{-\infty\}$.

(a) Show that $\mathcal{B}(\bar{\mathbb{R}}) = \big\{A \cup B : A \in \mathcal{B}(\mathbb{R}), B \subset \{-\infty,\infty\}\big\}$.

(b) Show, however, that the σ-algebra on $\bar{\mathbb{R}}$ generated by $\mathcal{B}(\mathbb{R})$ is given by $\sigma\langle\mathcal{B}(\mathbb{R})\rangle = \big\{A \cup B : A \in \mathcal{B}(\mathbb{R}), B = \{-\infty,\infty\}$ or $B = \emptyset\big\}$.

1.14 Let $\mathcal{A}_1, \mathcal{A}_2$ and $\mathcal{A}_3$ denote, respectively, the class of triangles, discs, and pentagons in $\mathbb{R}^2$. Show that $\sigma\langle\mathcal{A}_i\rangle \equiv \mathcal{B}(\mathbb{R}^2)$. Thus, the σ-algebra $\mathcal{B}(\mathbb{R}^2)$ (and similarly $\mathcal{B}(\mathbb{R}^k)$) can be generated by starting with various classes of sets of different shapes and geometry.

1.15 Let Ω be a nonempty set and $\mathcal{B}$ be a σ-algebra on Ω. Let $A \subset \Omega$ and $\mathcal{B}_A \equiv \{B \cap A : B \in \mathcal{B}\}$. Show that $\mathcal{B}_A$ is a σ-algebra on A. The σ-algebra $\mathcal{B}_A$ is called the *trace σ-algebra of $\mathcal{B}$ on A*.

1.16 Let $\Omega = \mathbb{R}$ and $\mathcal{F}$ be the collection of all finite unions of disjoint intervals of the form $(a,b] \cap \mathbb{R}$, $-\infty \le a < b \le \infty$. Show that $\mathcal{F}$ is an algebra but not a σ-algebra.

1.17 Let Ω be a nonempty set and $\{A_i\}_{i\in\mathbb{N}}$ be a sequence of subsets of Ω such that $A_{i+1} \subset A_i$ for all $i \in \mathbb{N}$. Verify that $\mathcal{A} = \{A_i : i \in \mathbb{N}\}$ is a π-system and also determine $\lambda\langle\mathcal{A}\rangle$, the λ-system generated by $\mathcal{A}$.

1.18 Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and $A_n = \{j : j \in \mathbb{N}, j \ge n\}, n \in \mathbb{N}$. Let μ be the counting measure on $(\Omega, \mathcal{F})$. Verify that $\lim_{n\to\infty} \mu(A_n) \neq \mu(\bigcap_{n\ge 1} A_n)$.

1.19 Let Ω be a nonempty set and let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a semialgebra. Let

$$\mathcal{A}(\mathcal{C}) \equiv \Big\{A : A = \bigcup_{i=1}^{k} B_i : B_i \in \mathcal{C}, i = 1, 2, \ldots, k, k \in \mathbb{N}\Big\}.$$

(a) Show that $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$.

(b) Show also that $\sigma\langle\mathcal{C}\rangle = \sigma\langle\mathcal{A}(\mathcal{C})\rangle$.

1.20 Let μ^* be as in (3.1) of Section 1.3. Verify (3.4)–(3.6).

(**Hint:** Fix $0 < \epsilon < \infty$. If $\mu^*(A_n) < \infty$ for all $n \in \mathbb{N}$, then find, for each n, a cover $\{A_{nj}\}_{j\ge1} \subset \mathcal{C}$ such that $\mu^*(A_n) \le \sum_{j=1}^{\infty} \mu(A_{nj}) + \frac{\epsilon}{2^n}$.)

1.21 Prove Proposition 1.3.1.

1.22 Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing. Let $(a,b], (a_n,b_n], n \in \mathbb{N}$ be intervals in $\mathbb{R}$ such that $(a,b] = \bigcup_{n\ge1}(a_n,b_n]$ and $\{(a_n,b_n] : n \ge 1\}$ are disjoint. Let $\mu_F(\cdot)$ be as in (3.8). Show that $\mu_F((a,b]) = \sum_{n=1}^{\infty} \mu_F((a_n,b_n])$ by completing the following steps:

(a) Let $G(x) \equiv F(x+)$ for all $x \in \mathbb{R}$ and let $G(\pm\infty) = F(\pm\infty)$. Verify that $G(\cdot)$ is nondecreasing and right continuous on $\mathbb{R}$ and that for any A in $\mathcal{C}$, $\mu_F(A) = \mu_G(A)$.

(b) In view of (a), assume w.l.o.g. that $F(\cdot)$ is right continuous. Show that for any $k \in \mathbb{N}$,

$$F(b) - F(a) \ge \sum_{i=1}^{k} (F(b_i) - F(a_i)),$$

and conclude that

$$F(b) - F(a) \geq \sum_{i=1}^{\infty} (F(b_i) - F(a_i)).$$

(c) To prove the reverse inequality, fix $\eta > 0$. Choose $c > a$ and $d_n > b_n$, $n \geq 1$ such that such that $F(c) - F(a) < \eta$, $[c, b] \subset \bigcup_{n \geq 1} (a_n, d_n)$ and $F(d_n) - F(b_n) < \eta/2^n$ for all $n \in \mathbb{N}$. Next, apply the Heine-Borel theorem to the interval $[c, b]$ and the open cover $\{(a_n, d_n)\}_{n \geq 1}$ and extract a finite cover $\{(a_i, d_i)\}_{i=1}^k$ for $[c, b]$. W.l.o.g., assume that $c \in (a_1, d_1)$ and $b \in (a_k, d_k)$. Now verify that

$$\begin{aligned} F(b) - F(a) &\leq \sum_{j=1}^{k} (F(b_j) - F(a_j)) + 2\eta \\ &\leq \sum_{j=1}^{\infty} (F(b_j) - F(a_j)) + 2\eta. \end{aligned}$$

1.23 Extend the above arguments to the case when $(a, b]$ and $(a_i, b_i]$, $i \geq 1$ are not necessarily bounded intervals.

1.24 Verify that $\mathcal{C}_2$, defined in (3.11), is a semialgebra.

1.25 (a) Verify that the limit in (3.13) exists.

(b) Extend the arguments in Problems 1.22 and 1.23 to verify that μ_F of (3.12) and (3.13) is a measure on $\mathcal{C}_2$.

1.26 Establish Theorem 1.3.6 by completing the following:

(a) Suppose first that $\nu(\Omega) < \infty$. Verify that $\mathcal{L} \equiv \{A : A \in \sigma\langle\mathcal{C}\rangle, \mu^*(A) = \nu(A)\}$ is a λ system and use the π-λ theorem.

(b) Extend the above to the σ-finite case.

1.27 Prove Corollary 1.3.5 for Lebesgue measure $m(\cdot)$.

1.28 Let F be a discrete distribution function, i.e., F is of the form

$$F(x) \equiv \sum_{j=1}^{\infty} a_j I(x_j \leq x), \quad x \in \mathbb{R},$$

where $0 < a_j < \infty$, $\sum_{j \geq 1} a_j = 1$, $x_j \in \mathbb{R}$, $j \geq 1$. Show that $\mathcal{M}_{\mu_F^*} = \mathcal{P}(\mathbb{R})$.

(**Hint:** Show that $\mu_F^*(A^c) = 0$, where $A \equiv \{x_j : j \geq 1\}$, and use the fact that for any $B \subset \mathbb{R}$, $B \cap A \in \mathcal{B}(\mathbb{R})$.)

1.29 Let

$$F(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ x & \text{for} \quad 0 \le x \le 1 \\ 1 & \text{for} \quad x > 0. \end{cases}$$

Show that $\mathcal{M}_{\mu_F^*} \equiv \{A : A \in \mathcal{P}(\mathbb{R}), A \cap [0,1] \in \mathcal{M}\}$, where $\mathcal{M}$ is the σ-algebra of Lebesgue measurable sets as in Definition 1.3.7.

1.30 Let $F(\cdot) = \frac{1}{2}\Phi(\cdot) + \frac{1}{2}F_P(\cdot)$ where $\Phi(\cdot)$ is the standard normal cdf, i.e., $\Phi(x) \equiv \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} Au$ and $F_P(x) \equiv \sum_{k=0}^{\infty} e^{-2} \frac{2^k}{k!} I_{(-\infty,x]}^{(k)}$, $x \in \mathbb{R}$. Let $F_1 = \Phi$, $F_2 = F_P$ and $F_3 = F$. Let $A_1 = (0,1)$, $A_2 = \{x : x \in \mathbb{R}, \sin x \in (0, \frac{1}{2})\}$, $A_3 = \{x :$ for some integers $a_0, a_1, \ldots, a_k$, $k < \infty$, $\sum_{i=0}^{k} a_i x^i = 0\}$, the set of all algebraic numbers. Compute $\mu_{F_i}(A_j)$, $1 \le i, j \le 3$.

1.31 Let μ be a measure on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$ where Ω is a nonempty set. Let μ^* be the outer measure generated by μ and let $\mathcal{M}_{\mu^*}$ be the σ-algebra of μ^*-measurable sets as defined in Theorem 1.3.3.

(a) Show that for all $A \subset \Omega$, there exists a $B \in \sigma\langle\mathcal{C}\rangle$ such that $A \subset B$ and $\mu^*(A) = \mu^*(B)$.

(**Hint:** If $\mu^*(A) = \infty$, take B to be Ω. If $\mu^*(A) < \infty$, use the definition of μ^* to show that for each $n \ge 1$, there exists $\{B_{nj}\}_{j \ge 1} \subset \mathcal{C}$ such that $A \subset B_n \equiv \bigcup_{j \ge 1} B_{nj}$, $\mu^*(A) \le \sum_{j=1}^{\infty} \mu(B_{nj}) < \mu^*(A) + \frac{1}{n}$. Take $B = \bigcap_{n \ge 1} B_n$.)

(b) Show that for all $A \in \mathcal{M}_{\mu^*}$ with $\mu^*(A) < \infty$, there exists $B \in \sigma\langle\mathcal{C}\rangle$ such that $A \subset B$ and $\mu^*(B \setminus A) = 0$.

(**Hint:** Use (a) and the relation $B = A \cup (B \setminus A)$ with A and $B \setminus A = B \cap A^c$ in $\mathcal{M}_{\mu^*}$.)

(c) Show that if μ is σ-finite (i.e., there exist sets $\Omega_n, n \ge 1$ in $\mathcal{C}$ with $\mu(\Omega_n) < \infty$ for all $n \ge 1$ and $\bigcup_{n \ge 1} \Omega_n = \Omega$), then in (b), the hypothesis that $\mu^*(A) < \infty$ can be dropped.

(**Hint:** Assume w.l.o.g. that $\{\Omega_n\}_{n \ge 1}$ are disjoint. Apply (b) to $\{A_n \equiv A \cap \Omega_n\}_{n \ge 1}$.)

(d) Show that if μ is σ-finite, then $A \in \mathcal{M}_{\mu^*}$ iff there exist sets $B_1, B_2 \in \sigma\langle\mathcal{C}\rangle$ such that $B_1 \subset A \subset B_2$ and $\mu^*(B_2 \setminus B_1) = 0$.

(**Hint:** Apply (c) to both A and A^c.)

This shows that $\mathcal{M}_{\mu^*}$ is the completion of $\sigma\langle\mathcal{C}\rangle$ w.r.t. μ^*.

1.32 (*An outline of a proof of Corollary 1.3.5*). Let $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ be a Lebesgue Stieltjes measure space generated by a right continuous and nondecreasing function $F : \mathbb{R} \to \mathbb{R}$.

(a) Show that $A \in \mathcal{M}_{\mu_F^*}$ iff there exist Borel sets B_1 and $B_2 \in \mathcal{B}(\mathbb{R})$ such that $B_1 \subset A \subset B_2$ and $\mu_F^*(B_2 \setminus B_1) = 0$.

(**Hint:** Take $\mathcal{C}$ to be the semialgebra $\mathcal{C} = \{(a,b] : -\infty \leq a \leq b < \infty\} \cup \{(b,\infty) : -\infty \leq b < \infty\}$ and apply Problem 1.31 (d).)

(b) Let $A \in \mathcal{M}_{\mu_F^*}$ with $\mu_F^*(A) < \infty$. Show that for any $\epsilon > 0$, there exist a finite number of bounded open intervals $I_j, j = 1, 2, \ldots, k$ such that $\mu_F^*(A \triangle \bigcup_{j=1}^k I_j) < \epsilon$.

(**Outline:** *Claim: For any $B \in \mathcal{C}$ with $\mu_F(B) < \infty$, there exists an open interval I such that $\mu_F^*(I \triangle B) < \epsilon$.*

To see this, note that if $B = (a,b], \ -\infty \leq a < b < \infty$, then one may choose $b' > b$ such that $F(b') - F(b) < \epsilon$. Now, with $I = (a,b')$, $\mu_F^*(I \triangle B) = \mu_F^*((b,b')) = \mu_F((b,b')) \leq F(b') - F(b) < \epsilon$. If $B = (b,\infty)$ and $\mu_F(B) < \infty$, then there exists $b' > b$ such that $F(\infty) - F(b'-) < \epsilon$. Hence, with $I = (b,b')$, $\mu_F^*(I \triangle B) = \mu_F^*([b',\infty)) = F(\infty) - F(b'-) < \epsilon$. This proves the claim.

Next, By Theorem 1.3.4, for all $\epsilon > 0$, there exist $B_1, B_2, \ldots, B_k \in \mathcal{C}$ such that $\mu_F^*(A \triangle \bigcup_{j=1}^k B_j) < \epsilon/2$. For each B_j, find I_j, a bounded open interval such that $\mu_F^*(B_j \triangle I_j) < \frac{\epsilon}{2^j}$. Since $(A_1 \cup A_2) \triangle (C_1 \cup C_2) \subset (A_1 \triangle C_1) \cup (A_2 \triangle C_2)$ for any $A_1, A_2, C_1, C_2 \subset \Omega$, it follows that

$$\mu_F^*\Big(\Big[\bigcup_{j=1}^k B_j\Big] \triangle \Big[\bigcup_{j=1}^k I_j\Big]\Big) < \sum_{j=1}^k \mu_F^*(B_j \triangle I_j) < \frac{\epsilon}{2}.$$

Hence, $\mu_F^*(A \triangle [\bigcup_{j=1}^k I_j]) < \epsilon$.)

(c) Let $A \in \mathcal{M}_{\mu_F^*}$ with $\mu_F^*(A) < \infty$. Show that for every $\epsilon > 0$, there exists an open set O such that $A \subset O$ and $\mu_F^*(O \setminus A) < \epsilon$.

(**Hint:** By definition of μ_F^*, for every $\epsilon > 0$, there exist $\{B_j\}_{j\geq 1} \subset \mathcal{C}$ such that $A \subset \bigcup_{j\geq 1} B_j$ and $\mu_F^*(A) \leq \sum_{j=1}^\infty \mu_F(B_j) \leq \mu_F^*(A) + \epsilon$. Now as in (b), there exist open intervals I_j such that $B_j \subset I_j$ and $\mu_F^*(I_j \setminus B_j) < \epsilon/2^j$ for all $j \geq 1$. Then $A \subset \bigcup_{j=1}^\infty B_j \subset \bigcup_{j=1}^\infty I_j \equiv O$. Also, $\mu_F^*(O) = \mu_F^*(A) + \mu_F^*(O \setminus A) \Rightarrow \mu_F^*(O \setminus A) = \mu_F^*(O) - \mu^*(A) < 2\epsilon$ (since $\mu^*(O) \leq \sum_{j=1}^\infty \mu^*(I_j) = \sum_{j=1}^\infty \mu_F^*(B_j) + \epsilon \leq \mu_F^*(A) + 2\epsilon < \infty$).)

(d) Extend (c) to all $A \in \mathcal{M}_{\mu_F^*}$.

(**Hint:** Let $A_i = A \cap [i, i+1]$, $i \in \mathbb{Z}$. Apply (c) to A_i with $\epsilon_i = \frac{\epsilon}{2^{|i|+1}}$ and take unions.)

(e) Show that for all $A \in \mathcal{M}_{\mu_F^*}$ and for all $\epsilon > 0$, there exist a closed set C and an open set O such that $C \subset A \subset O$ and

$\mu_F^*(O \setminus A) \le \epsilon$, $\mu^*(A \setminus C) < \epsilon$.

(**Hint:** Apply (d) to A and A^c.)

(f) Show that for all $A \in \mathcal{M}_{\mu_F^*}$ with $\mu_F^*(A) < \infty$ and for all $\epsilon > 0$, there exist a closed and bounded set $F \subset A$ such that $\mu_F^*(A \setminus F) < \epsilon$ and an open set O with $A \subset O$ such that $\mu_F^*(O \setminus A) < \epsilon$.

(**Hint:** Apply (d) to $A \cap [-M, M]$ where M is chosen so that $\mu_F^*(A \cap [-M, M]^c) < \epsilon$. Why is this possible?)

Remark: Thus for all $A \in \mathcal{M}_{\mu_F^*}$ with $\mu_F^*(A) < \infty$ and for all $\epsilon > 0$, there exist a compact set $K \subset A$ and an open set $O \supset A$ such that $\mu_F^*(A \setminus K) < \epsilon$ and $\mu_F^*(O \setminus A) < \epsilon$. The first property is called *inner regularity of* μ_F^* and the second property is called *outer regularity* of μ_F^*.

(g) Show that for all $A \in \mathcal{M}_{\mu_F^*}$ with $\mu_F^*(A) < \infty$ and for all $\epsilon > 0$, there exists a continuous function g_ϵ with compact support (i.e., $g_\epsilon(x)$ is zero for $|x|$ large) such that

$$\mu_F^*(A \bigtriangleup \; g_\epsilon^{-1}\{1\}) < \epsilon.$$

(**Hint:** For any bounded open interval (a, b), let $\eta > 0$ be such that $\mu_F((a, a + \eta]) + \mu_F([b - \eta, b)) < \epsilon$. Next define

$$g_\epsilon(x) = \begin{cases} 1 & \text{if} \quad a + \eta \le x \le b - \eta \\ 0 & \text{if} \quad x \notin (a, b) \\ \text{linear} & \text{over} \quad [a, a + \eta] \cup [b - \eta, b]. \end{cases}$$

Then g_ϵ is continuous with compact support. Also, $g^{-1}\{1\} = [a + \eta, b - \eta]$ and $(a, b) \bigtriangleup g^{-1}\{1\} = (a, a + \eta) \cup (b - \eta, b)$. So $\mu_F\{(a, b) \bigtriangleup g_\epsilon^{-1}(1)\}| < \epsilon$, proving the claim for $A = (a, b)$. The general case follows from (b).)

(h) Show that for all $A \in \mathcal{M}_{\mu_F^*}$ and for all $\epsilon > 0$, there exists a continuous function g_ϵ (not necessarily with compact support) such that $\mu_F^*(A \bigtriangleup g_\epsilon^{-1}\{1\}) < \epsilon$ (i.e., drop the condition $\mu_F^*(A) < \infty$).

(**Hint:** Let $A_k = A \cap [k, k + 1]$, $k \in \mathbb{Z}$. Find $g_k : \mathbb{R} \to \mathbb{R}$ continuous with support in $\left(k - \frac{1}{8}, k + \frac{9}{8}\right)$ such that $\mu_F^*(I_{A_k} \neq g_k) < \frac{\epsilon}{2^{|k|+1}}$. Let $g = \sum_{k \in \mathbb{Z}} g_k$. Note that for any $x \in \mathbb{R}$, at most two $g_k(x) \neq 0$ and so g is continuous. Also, $\mu_F^*(I_A \neq g) \le \sum_{k \in \mathbb{Z}} \mu_F^*(I_{A_k} \neq g_k) < \epsilon$.)

1.33 Let C be the Cantor set in [0,1] as defined in Section 1.3.2.

(a) Show that

$$C = \Big\{x : x = \sum_{i=1}^{\infty} \frac{a_i}{3^i}, \; a_i \in \{0, 2\}\Big\}.$$

and hence that C is uncountable.

(b) Show that

$$C + C \equiv \{x + y : x, y \in C\} = [0, 2].$$